

ICLAB 2024 Autumn

LEC 11 : Cell-Based APR Design Flow

Lecturer: Bang-Yuan Xiao



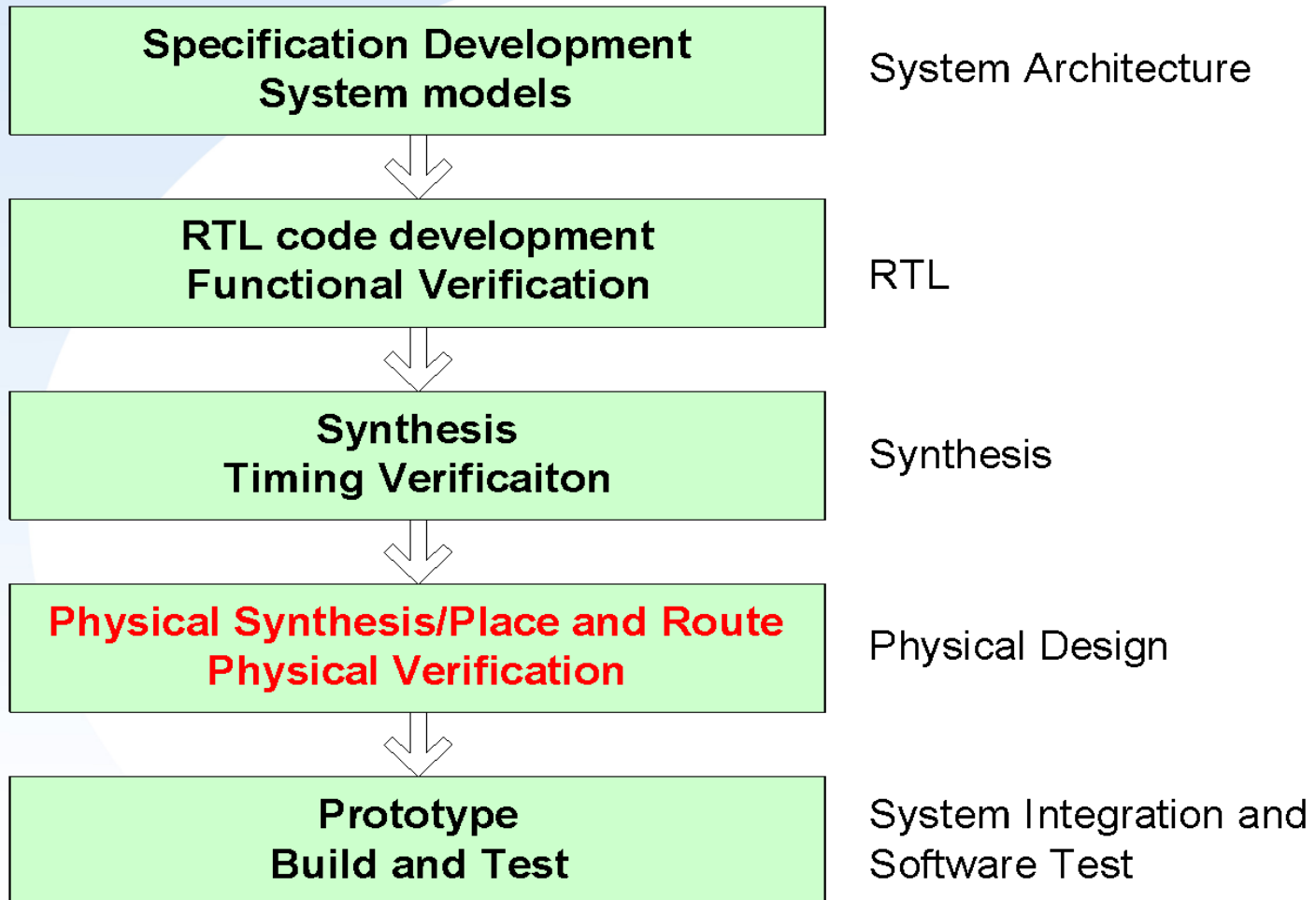
Outline

- ✓ **Section 1- What's APR**
- ✓ **Section 2- SOP of APR**



Introduction

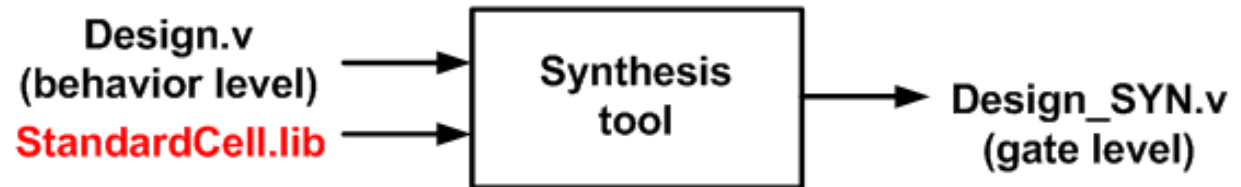
✓ Cell-based Design Flow



Review Cell-based Design Flow

✓ RTL coding

✓ Synthesis



✓ Physical Implementation

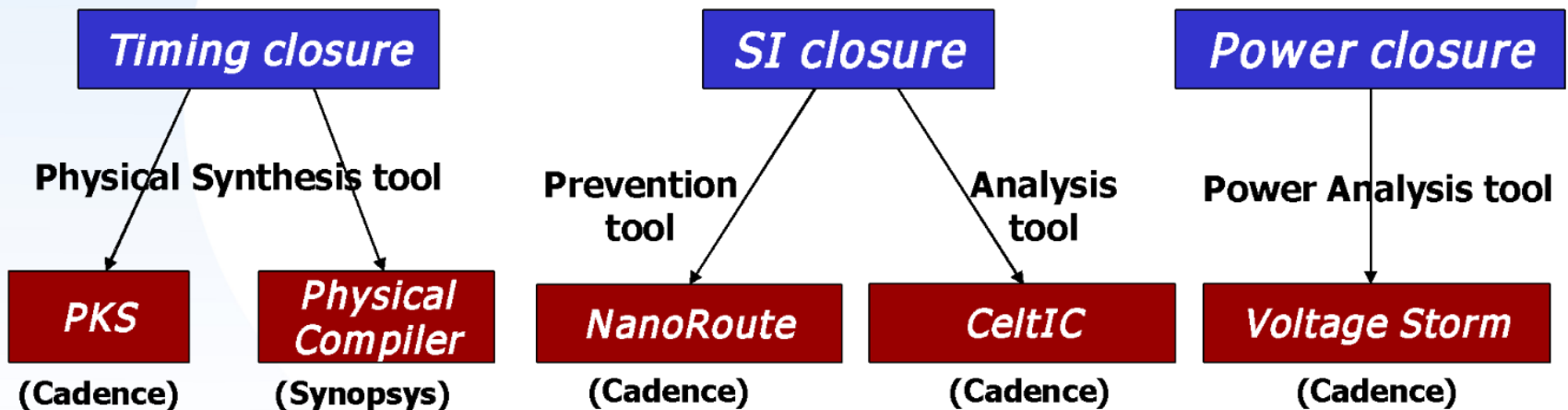


IC Compiler · Innovus

Wiring Problem

✓ New problems due to large wire resistance and capacitance

- Timing closure
- Signal Integrity(SI) closure(crosstalk, ...)
- Power closure (IR drop, ...)



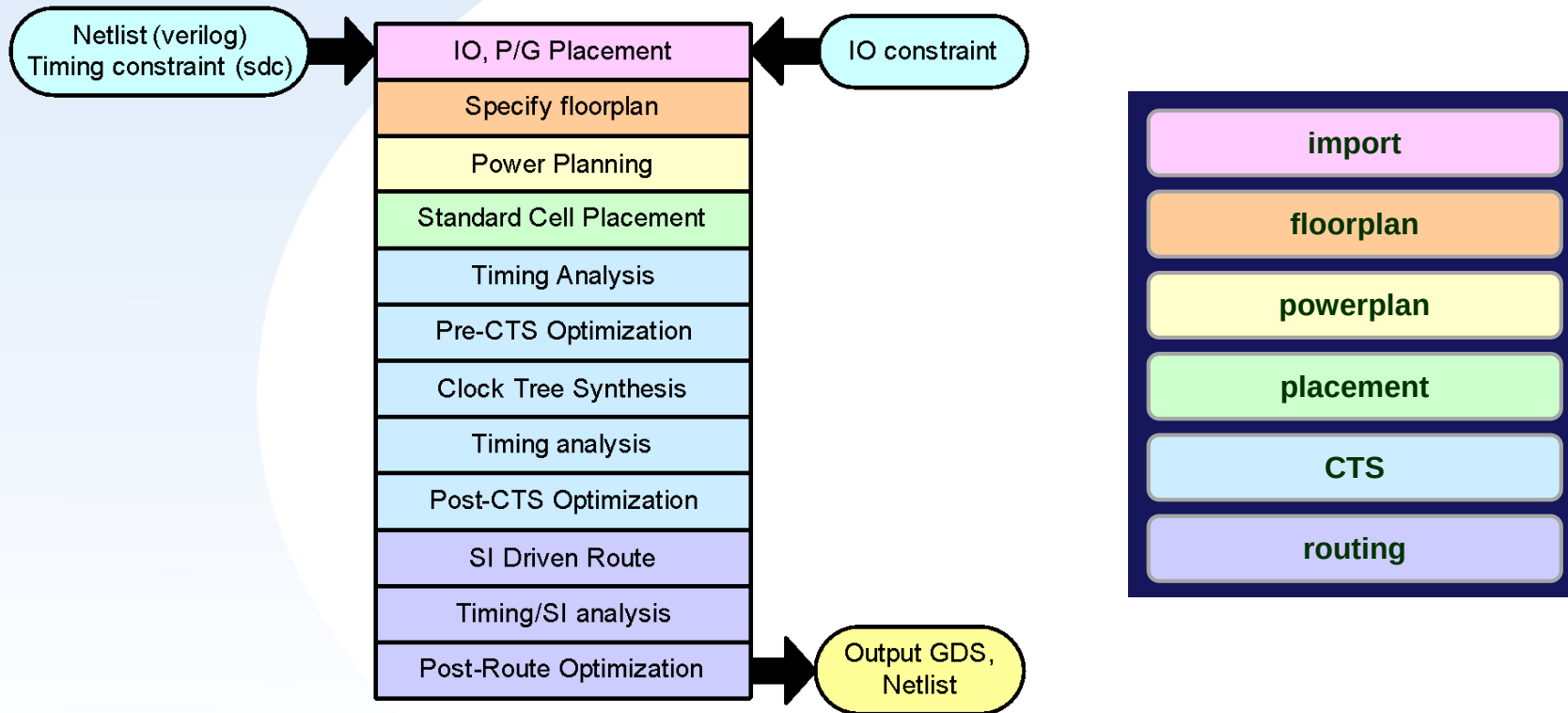
Outline

- ✓ Section 1- What's APR
- ✓ Section 2- SOP of APR



APR Flow

Innovus Design Flow



1. Data Preparation

✓ Library files

- Timing libraries (LIB)
 - slow.lib
 - fast.lib
 - umc18io3v5v_slow.lib
 - umc18io3v5v_fast.lib
 - HardMacro_slow.lib
 - HardMacro_fast.lib
- Physical libraries (LEF)
 - umc18_6lm.lef
 - umc18io3v5v_6lm.lef
 - umc18_6lm_antenna.lef
 - HardMacro.vclef
- RC extraction
 - umc18_1p6m.captbl
 - RCGen.tch
- CeltIC libraries
 - slow.cdb
 - fast.cdb

- GDSII layout
 - umc18.gds2
 - umc18io3v5v_61m.gds2
 - HardMacro.gds2

✓ Timing constraint files (User Data)

- Generate timing constraint files from Design Compiler
 - dc_shell-t> **write_sdc CHIP.sdc**

✓ Design netlist files (User Data)

- Synthesized design netlist
 - **core_syn.v**
- Chip design netlist (combine core_syn.v&chip_shell.v)
 - **CHIP_SYN.v**
- IO pad location file
 - **CHIP.io**



1. Data Preparation – Library Files

✓ LIB Library : timing information

- Calculate cell delay according to different combinations of input slew rate and output loading capacitance
- Using SignalStorm Library Characterizer

✓ LEF Library : process and cell information

- Metal layer information, routing pitch, default direction
- Cell size, pin position, pin metal layer

✓ RC Extraction

- RC extraction for further power analysis, simulation etc

✓ CeltIC Library

- cdB model (noise model for each cell which will be use in SI optimization phase)

✓ GDSII

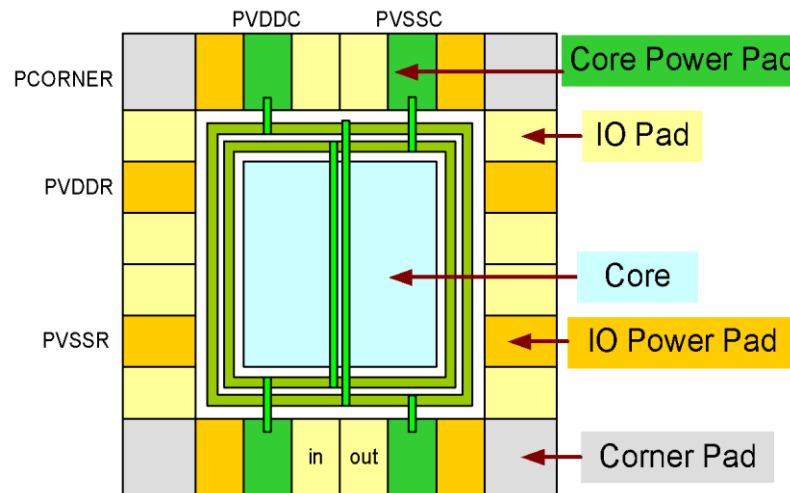
- planar geometric shapes and other information about the layout in hierarchical form



1. Data Preparation - Design Netlist

✓ Prepare chip design netlist: **CHIP_SYN.v**

- Add pad netlist to synthesized design netlist
- Combine *designName_SYN.v* and *CHIP_SHELL.v*
 - `cat CHIP_SHELL.v CORE_SYN.v > CHIP_SYN.v`
- **CHIP_SHELL.v** contains I/O, I/O power, and core power
 - Note: Refer to pad cell library datasheet
 - [evince ~iclabta01/umc018/Doc/umc18io3v5v.pdf &](#)



1. Data Preparation - Design Netlist

✓ Core/IO power pad selection (UMC 0.18)

— Pad cell categories

Special Cells

Core power pads: PVDDC, PVSSC

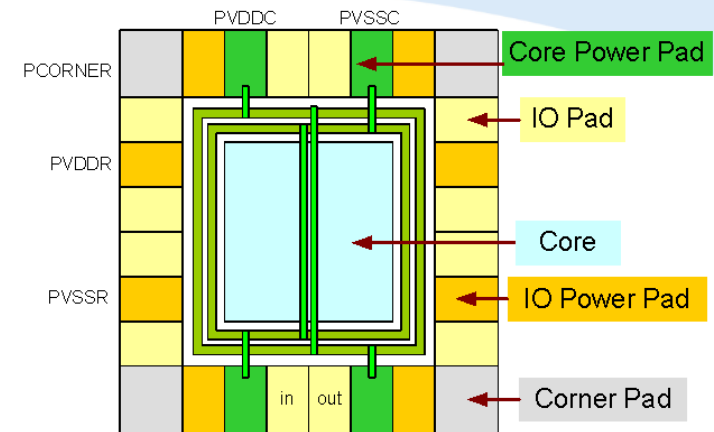
IO power pads: PVDDR, PVSSR

I/O filler pads: PFILL, ...

I/O Cells

P2A	P4A	P8A	P16A	P24A
		P8B	P16B	P24B
P2C	P4C	P8C	P16C	P24C

R



```
PVDDC VDDC0 ();
PVSSC GND0 ();
P8C I_CLK [ .Y(C_clk), .P(c1k), .A(1'b0), .ODEN(1'b0),
| .OCEN(1'b0), .PU(1'b1), .PD(1'b0), .CEN(1'b0), .CSEN(1'b1) ];
```

— Core power pad

- One set core power pad (PVDDC along with PVSSC) can provide 40~50mA current

— IO power pad

- One set IO power pad (PVDDR along with PVSSR) can provide the power for
 - ◆ 3~4 output pads/PG pad set
 - ◆ 6~8 input pads pads/PG pad set

1. Data Preparation – I/O Files

✓ Prepare IO pad location file: **CHIP.io**

- Edit IO pad location file
 - **CHIP.io**
- Syntax
 - Pad: `pad_instance_name direction [pad_type]`
- Categories

Field	Usage	
pad_instance_name	Corresponding instance name of pad in the chip design netlist	
direction	Specify pad location in the floorplan	S (southern pads)
		E (eastern pads)
		W (western pads)
		N (northern pads)
pad_type	Specify pad type (only for pad filler and corner pad)	PFILL (pad fillers)
		PCORNER (corner pads)



1. Data Preparation – I/O Files

✓ Example of IO pad location file

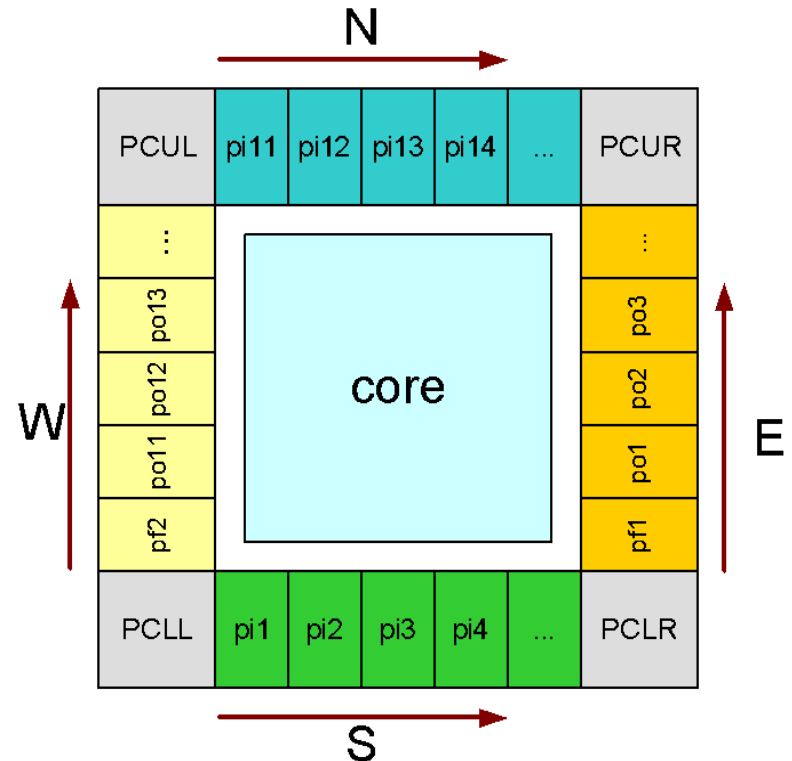
CHIP_SYN.v

```
PVDDR pi11 ();  
PVSSR pi12 ();
```

CHIP.io

Version: 2

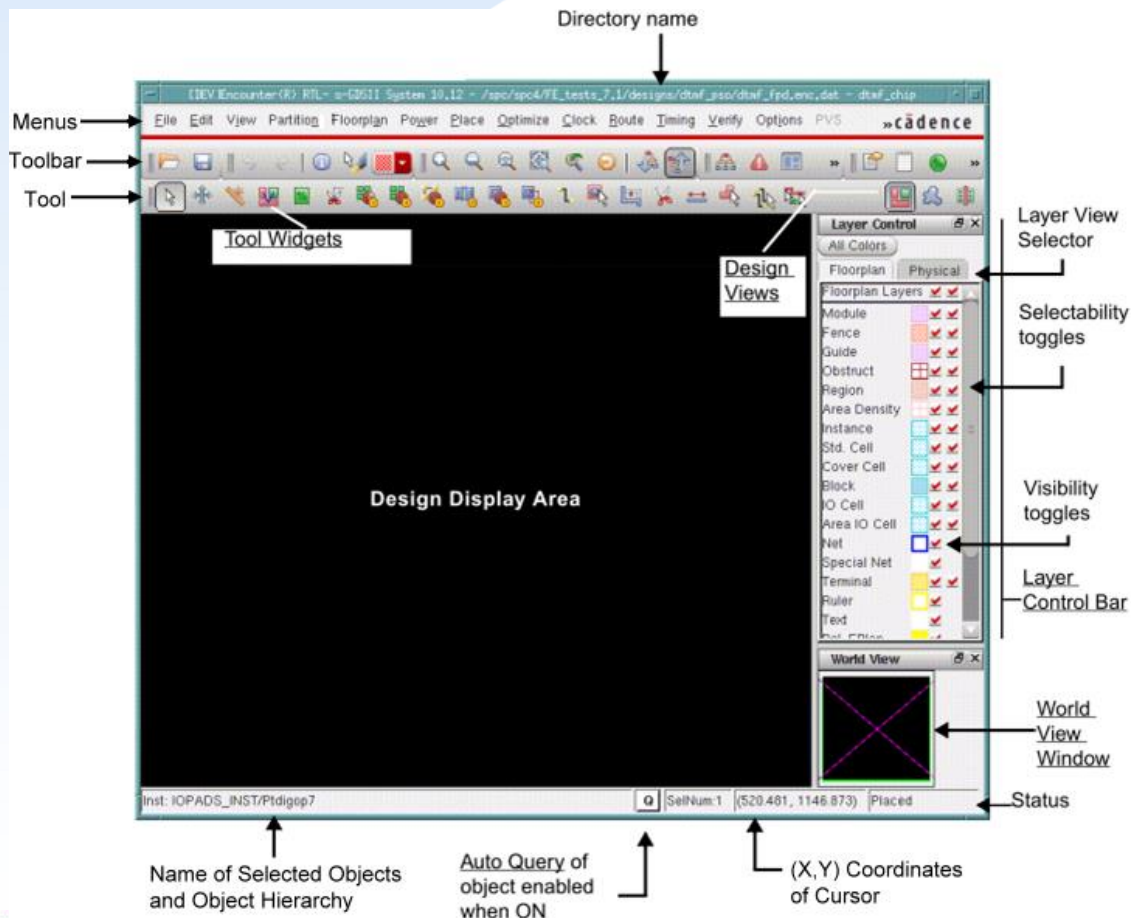
Pad: pi1	S	
Pad: pi2	S	
Pad: pi3	S	
Pad: pi4	S	
.....		
Pad: pf1	E	PFILL
Pad: po1	E	
Pad: po2	E	
Pad: po3	E	
.....		
Pad: pi11	N	
Pad: pi12	N	
Pad: pi13	N	
Pad: pi14	N	
.....		
Pad: pf2	W	PFILL
Pad: po11	W	
Pad: po12	W	
Pad: po13	W	
.....		
Pad: PCLL	SW	PCORNER
Pad: PCLR	SE	PCORNER
Pad: PCUL	NW	PCORNER
Pad: PCUR	NE	PCORNER



2. Importing Design – Invoke Innovus

✓ **Command to start an Innovus session**

– UNIX% **innovus**



Note:

1. **Do not use background execution!!!**
(The terminal become the interface of command input while running soc innovus)

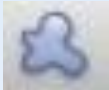
2. Log file name:
innovus.log
innovus.log1
innovus.log2
...

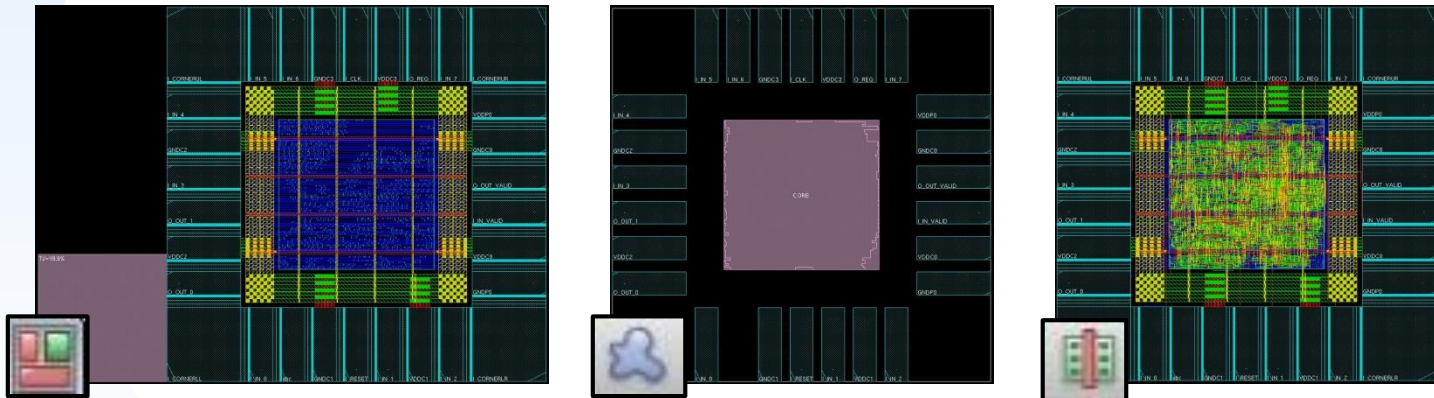
3. Command file name:
innovus.cmd
innovus.cmd1
innovus.cmd2
...



2. Importing Design – Innovus Views

✓ Design Views

-  Floorplan View: displays the hierarchical module and block guides, connection flight lines and floorplan objects
-  Amoeba View: display the outline of modules after placement
-  Placement View: display the detailed placements of cells.



✓ Set design view

- innovus #> [setDrawMode fplan](#)

2. Importing Design

import

floorplan

powerplan

placement

CTS

routing

✓ Import design files into Innovus environment

Design Edit Synthesis Partition Floorplan Power Place Clock Route Timing SI Verify Tools

Design Import

- Netlist:
By Verilog, and specifies the top cell by user
- Technology/Physical Libraries:
You must specify the technology LEF file first, then specify the standard cell LEF and block LEF. (Use LEF flow here)
- Floorplan
Specifies the I/O files to import.

The screenshot shows the 'Design Import' dialog box with the following sections:

- Netlist:** Radio buttons for Verilog (selected) and OA. Fields for Files (CHIP_SYN.v), Top Cell (Auto Assign / By User), Library, Cell, and View.
- Technology/Physical Libraries:** Radio buttons for OA and LEF Files (selected). Fields for Reference Libraries, Abstract View Names, Layout View Names, and a list of LEF files (8_6lm.lef, umc18_6lm_antenna.lef, umc18io3v5v_6lm.lef, RA1SH1.vclef).
- Floorplan:** IO Assignment File (CHIP.io).
- Power:** Power Nets (VDD), Ground Nets (GND), and CPF File.
- Analysis Configuration:** MMMC View Definition File (CHIP_mmmc.view) and a 'Create Analysis Configuration ...' button.

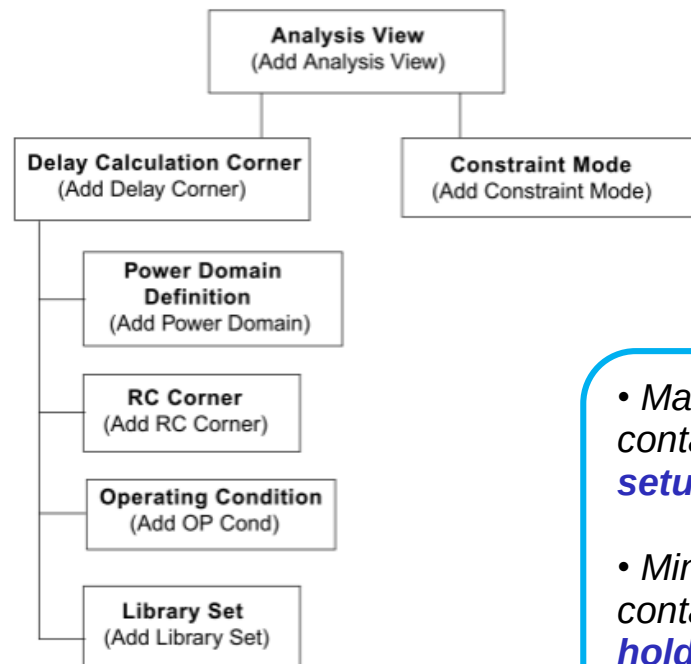
At the bottom are buttons for OK, Save... (circled in red), Load..., Cancel, and Help.



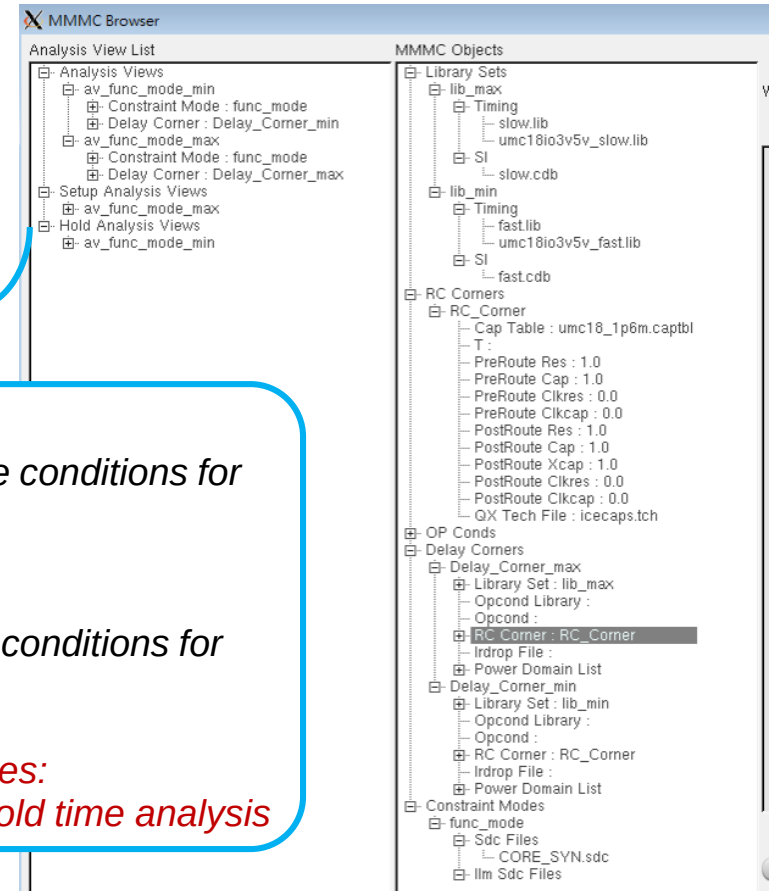
2. Importing Design – MMMC

✓ Multi-mode multi-corner (MMMC) timing analysis and optimization

- Press “Create Analysis Configuration” tab
- In MMMC environment you must specify the lower-level information first.



- *Max Timing Libraries:*
containing the worst-case conditions for **setup-time** analysis
- *Min Timing Libraries:*
containing the best-case conditions for **hold-time** analysis
- *Common Timing Libraries:*
used in both setup and hold time analysis



import

floorplan

powerplan

placement

CTS

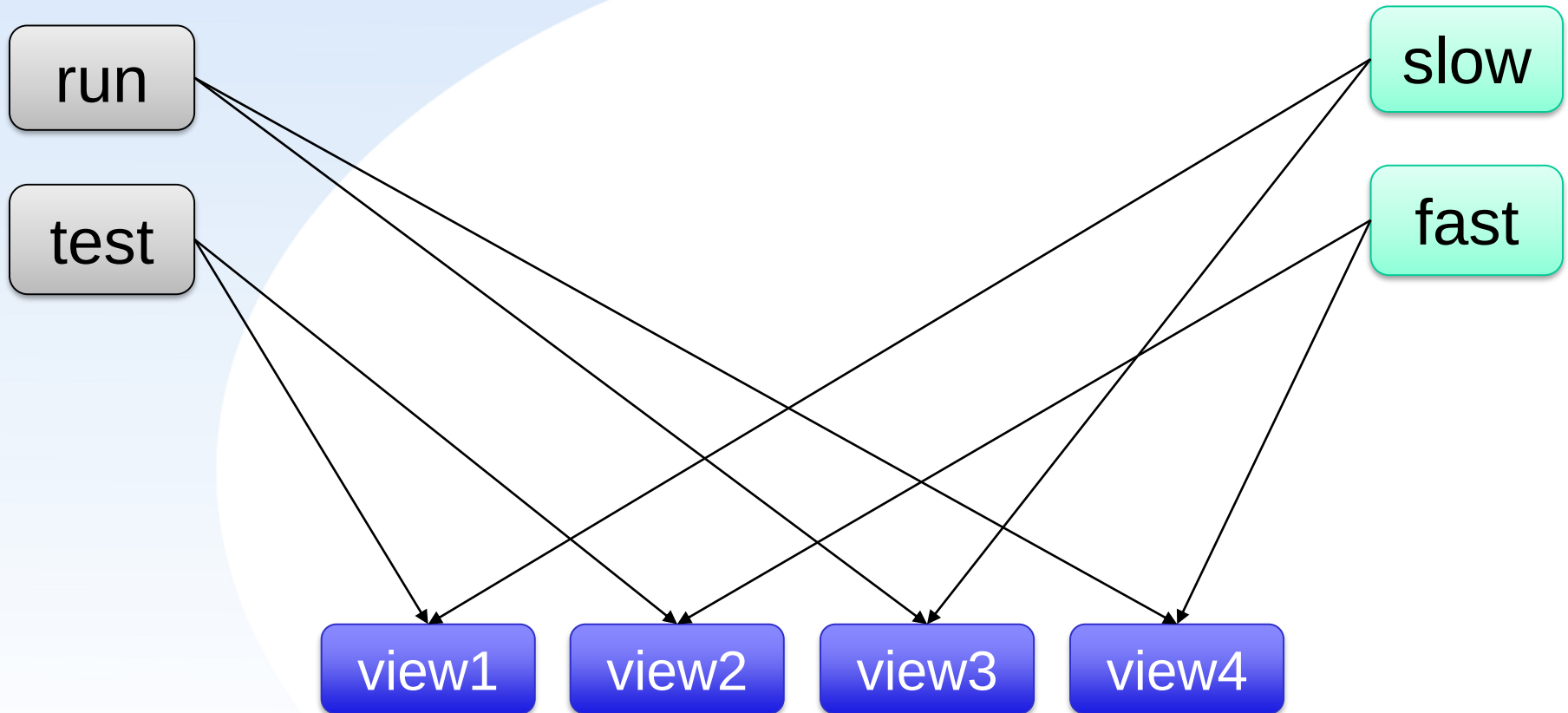
routing



2. Importing Design – MMMC

Multi-mode

Multi-corner



2. Importing Design - Result

import

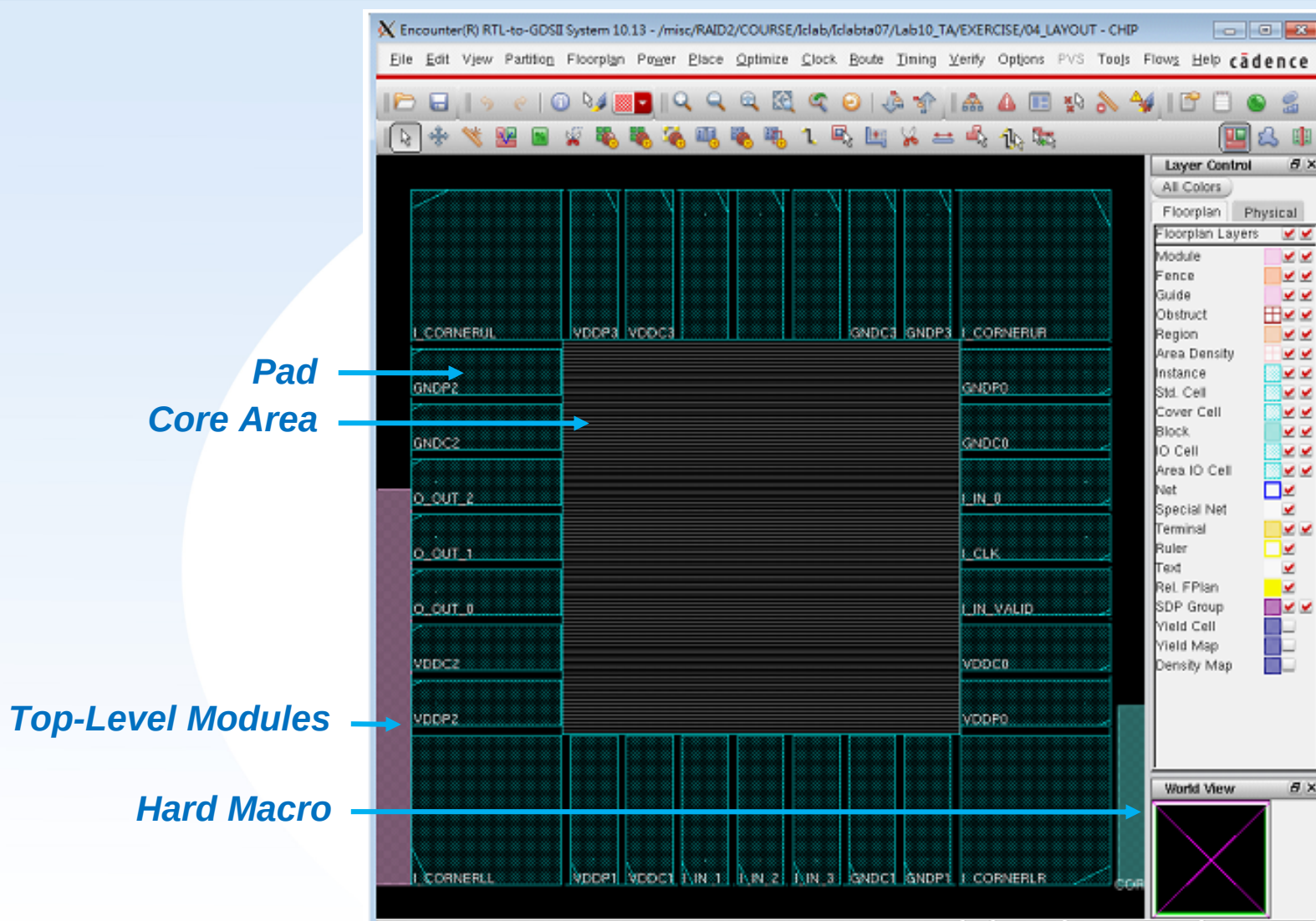
floorplan

powerplan

placement

CTS

routing



3. Floor Planning – Intro.

import

floorplan

powerplan

placement

CTS

routing

✓ Calculating core size, width and height

- When calculating core size of standard cells, the core utilization must be decided first. Usually the core utilization is higher than **70%**
- The core size can be calculated as follows

$$\text{Core Size of Standard Cell} = \frac{\text{standard cell area}}{\text{core utilization}}$$

- The recommended core shape is a square, i.e. Core Aspect Ratio = 1. Hence the width and height can be calculated as

$$\text{Width} = \text{Height} = \sqrt{\text{Core Size of Standard Cell}}$$

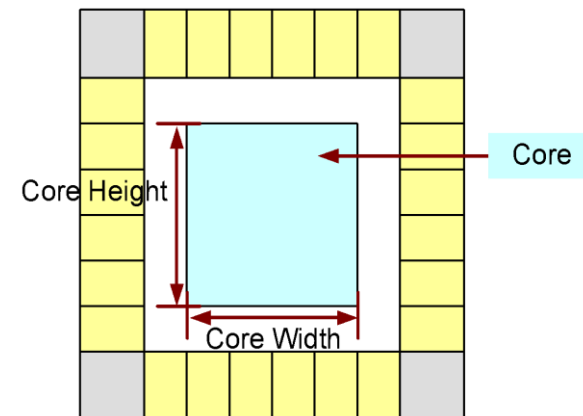
- Note: because stripes and macros will be added, width and height are usually set larger than the value calculated above

– E.g.

- Standard cell area = 2,000,000
- Core utilization demanded = 85%
- No macros

$$\text{Core Size of Standard Cell} = \frac{2,000,000}{0.85} = 2,352,941$$

$$\text{Width} = \text{Height} = \sqrt{2,352,941} = 1534$$



3. Floor Planning

import

floorplan

powerplan

placement

CTS

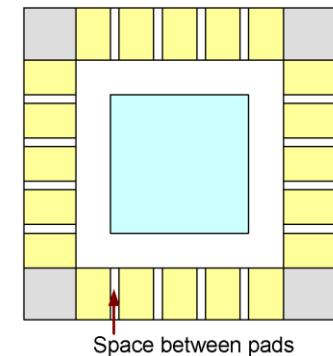
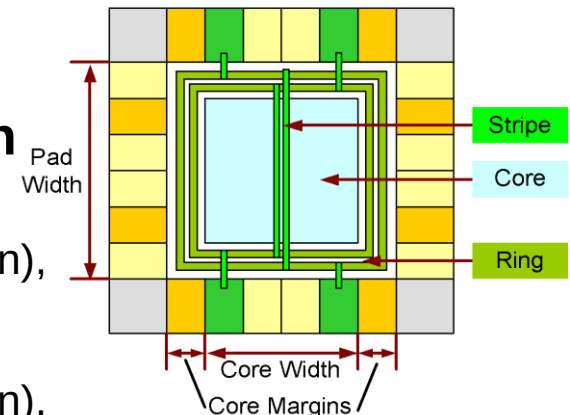
routing

✓ Core margins

- When setup the floorplan, remember to leave enough space for Power/Ground (P/G) ring
 - Note: the width needed for P/G ring will be discussed in power planning

✓ Core limited design or Pad limited design

- Die size determination
 - When $\text{pad width} > (\text{core width} + \text{core margin})$, die size is decided by pads. And it is called pad limited design
 - When $\text{pad width} < (\text{core width} + \text{core margin})$, die size is decided by core. And it is called core limited design
- Adding pad filler
 - Provide power to pad
 - There should be no spacing between pads. Pad filler is necessary for **core limited design**



Core limited design



3. Floor Planning

import

floorplan

powerplan

placement

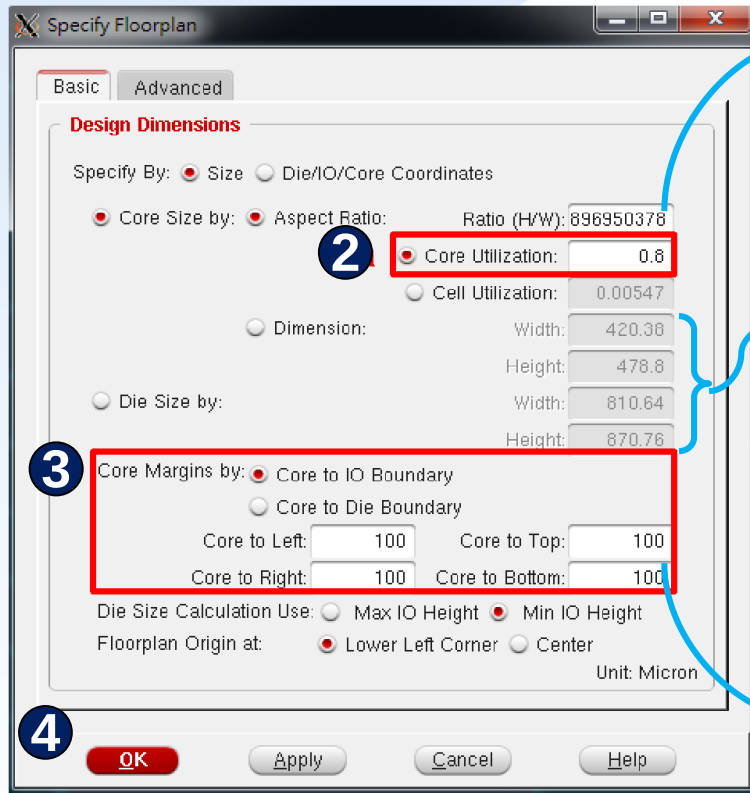
CTS

routing

Design Edit Synthesis Partition **Floorplan** Power Place Clock Route Timing SI Verify Tools

1

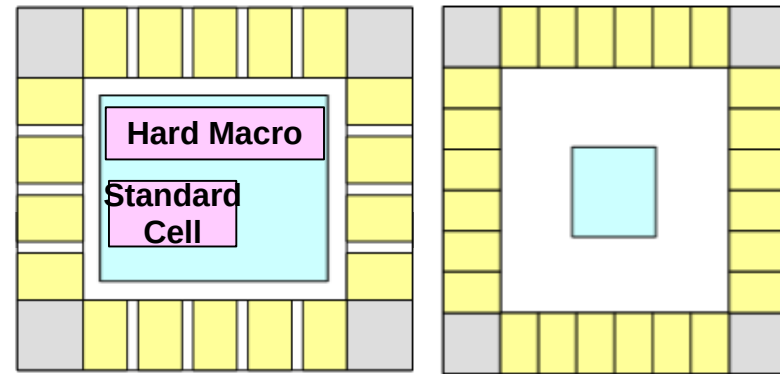
Specify Floorplan



The image shows the 'Specify Floorplan' dialog box with the 'Basic' tab selected. It contains several sections: 'Design Dimensions' with 'Specify By' set to 'Size', 'Core Size by' set to 'Aspect Ratio' with a ratio of 896950378, 'Core Utilization' set to 0.8, 'Cell Utilization' set to 0.00547, 'Dimension' with width 420.38 and height 478.8, and 'Die Size by' with width 810.64 and height 870.76. 'Core Margins by' is set to 'Core to IO Boundary' with margins of 100 on all sides. 'Die Size Calculation Use' is set to 'Min IO Height' and 'Floorplan Origin at' is set to 'Lower Left Corner'. The unit is 'Micron'. The dialog has 'OK', 'Apply', 'Cancel', and 'Help' buttons at the bottom.

Some space are remained for routing.

Set a demanded value of width and height, when Hard Macro takes most of the place.



• Core Limited Design • Pad Limited Design

*Some space are remained for power rings.
(design dependent)*

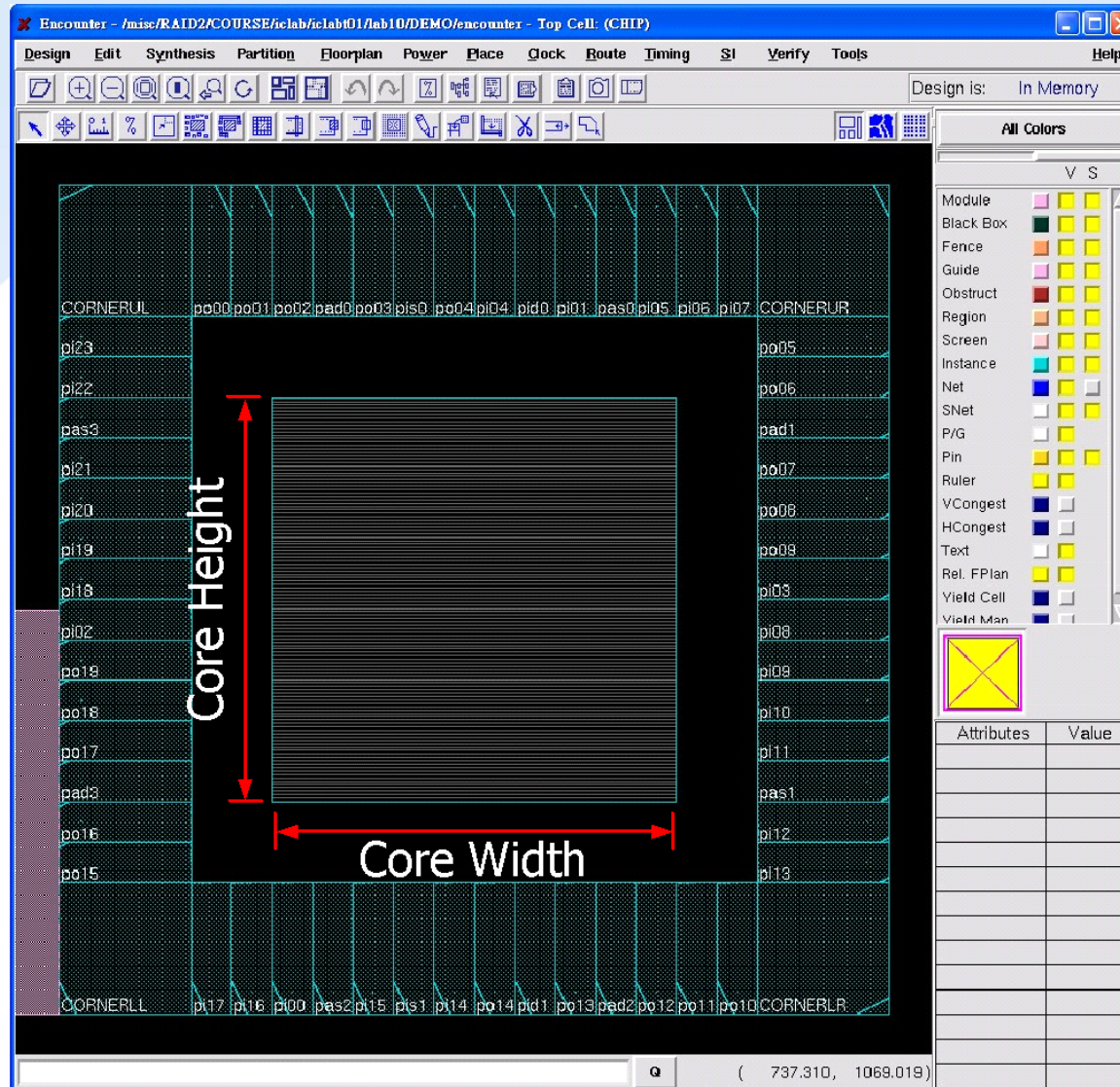
✓ Text Command (Specify core width, height and core margin)

– innovus #> floorPlan -s coreW coreH coreToLeft coreToBottomcoreToRight coreToTop



3. Floor Planning – Result

✓ Result



import

floorplan

powerplan

placement

CTS

routing



3. Floor Planning – Intro.

import

floorplan

powerplan

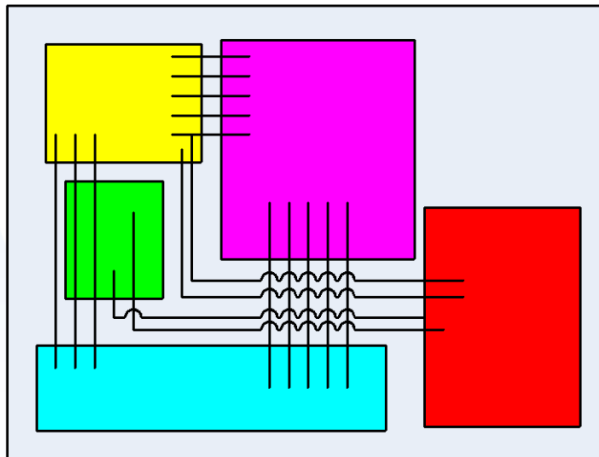
placement

CTS

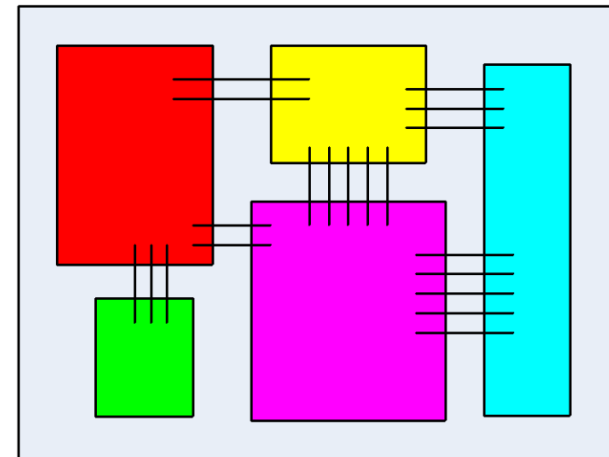
routing

✓ Floorplan Purposes

- Develop early physical layout to ensure design objective can be achieved
 - Minimum area for low cost
 - Minimum congestion for design routable
 - Estimate parasitic for delay calculation
 - Analysis power for reliability
- Gain early visibility into implementation issues
- Different floorplan, different performance



Bad floorplan



Good floorplan

3. Floor Planning with Hard Macro

import

floorplan

powerplan

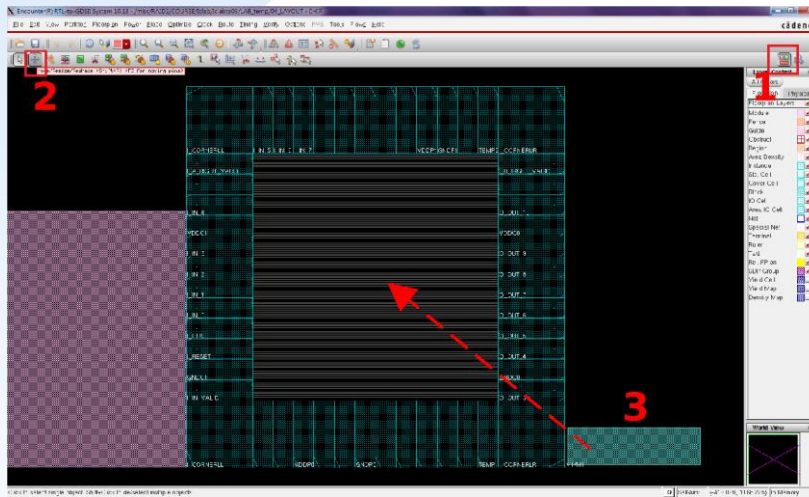
placement

CTS

routing

✓ Macro Floorplanning

- Move macro blocks to proper position
- Place macro around the corner to improve routability



Block place issue:

- power issue
- noise issue
- route issue

✓ In practice, if there are many hard macro, try to use automatic floorplan and the following tools

- Floorplan → Relative Floorplan → Define array constraint
- Floorplan → Placement toolbox



4. Power Planning – Intro.

import

floorplan

powerplan

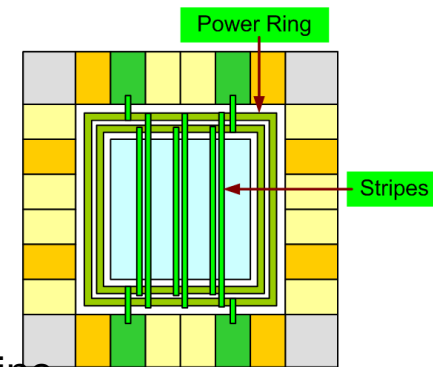
placement

CTS

routing

✓ Power issue

- Metal migration (also known as electro-migration)
 - Under **high currents**, electron collisions with metal grains cause the metal to move. The metal wire may be **open circuit** or **short circuit**.
 - Prevention: sizing power supply lines to ensure that the chip does not fail
 - Experience: make current density of power ring $< 1\text{mA}/\mu\text{m}$
- IR drop
 - IR drop is the problem of voltage drop of the power and ground due to high current flowing through the power-ground resistive network
 - When there are excessive voltage drops in the power network or voltage rises in the ground network, the device will run at **slower speed**
 - IR drop can cause the chip to fail due to
 - ◆ Performance (circuit running slower than specification)
 - ◆ Functionality problem (setup or hold violations)
 - ◆ Unreliable operation (less noise margin)
 - ◆ Power consumption (leakage power)
 - ◆ Latch up
 - Prevention: adding stripes to avoid IR drop on cell's power line



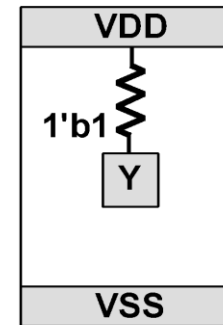
4.1. Power Planning

- Connect/Define Global Nets

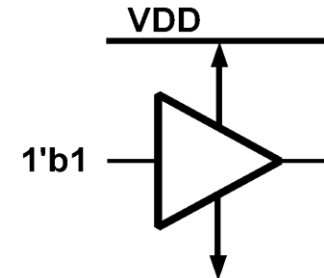
✓ 4.1. Connect/Define Global Nets

– Complete the form

- Add power nets and pins connection list
- Add ground nets and pins connection list
- Tie High: Connect all the 1'b1 to VDD
- Tie Low: Connect all the 1'b0 to GND



Tie High Cell



– After Applying, Only warnings of IO pads and padfiller are allowed

– Text Command

- innovus #> clearGlobalNets
- innovus #> globalNetConnect VDD -type pgpin -pin VDD -inst *
- innovus #> globalNetConnect VDD -type net -net VDD
- innovus #> globalNetConnect VDD -type tiehi -inst *
- innovus #> globalNetConnect GND -type pgpin -pin GND -inst *
- innovus #> globalNetConnect GND -type net -net GND
- innovus #> globalNetConnect GND -type tielo -inst *
- innovus #> globalNetConnect GND -type pgpin -pin VSS -inst *

4.1. Power Planning

- Connect/Define Global Nets

import
floorplan
powerplan
placement
CTS
routing

Design Edit Synthesis Partition Floorplan **Power** Place Clock Route Timing SI Verify Tools

1

Connect Global Nets...

Create 6 global nets connection list

2 Connect ♦ Pins: *VDD*
To Global Net: *VDD*

3 Connect ♦ Net Basename: *VDD*
To Global Net: *VDD*

4 Connect ♦ Tie High
To Global Net: *VDD*

5 Connect ♦ Pins: *GND*
To Global Net: *GND*

6 Connect ♦ Net Basename: *GND*
To Global Net: *GND*

7 Connect ♦ Tie Low
To Global Net: *GND*

Connection List

- VDD:PIN:*, VDD:All
- VDD:NET:VDD:All
- VDD:TIEHI:*, VDD:All
- GND:PIN:*, GND:All
- GND:NET:GND:All
- GND:TIELO:*, GND:All

Power Ground Connection

Connect

- ☒ Pin
- ☐ Tie High
- ☐ Tie Low

Instance Basename: *

Pin Name(s):

☐ Net Basename:

Scope

- ☐ Single Instance:- ☒ Under Module:- ☐ Under Power Domain:- ☐ Under Region: llx: 0.0 lly: 0.0 urx: 0.0 ury: 0.0
- ☒ Apply All

To Global Net:

☐ Override prior connection

☐ Verbose Output

Add to List Update Delete

Apply Check Reset Cancel Help



4. Power Planning – RING

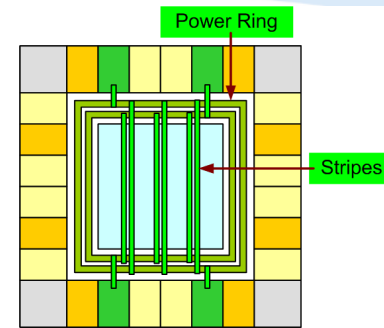
✓ Calculate power/ground ring width

– Experience

- Gate count = 70 k
- 4000 Flip-Flops
- 80% FF with dynamic gated clock
- Current needed = **0.2mA/MHz**
 - ◆ Note: the value should multiply with **1.8~2** for no gated design

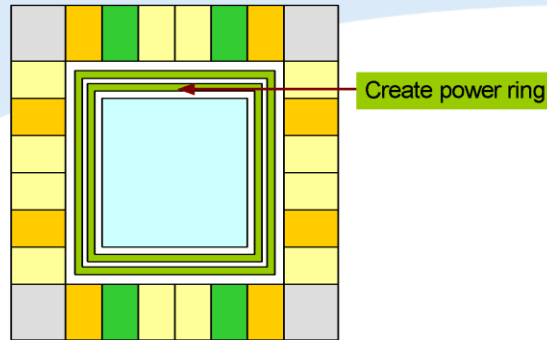
– Example:

- Gate count = 200 k
- No gated clock
- Clock frequency = 20 MHz
- Current needed = $(200k/70k) * 0.2 \text{ mA/MHz} * 20\text{MHz} * 2 = 22.86 \text{ mA}$
- Current density < $1\text{mA}/\mu\text{m}$
- The Width of P/G Ring > $22.86 \mu\text{m}$
- In order to avoid the slot rule of wide metal, the largest width is $20 \mu\text{m}$ (process dependent)
- Use two set P/G ring for this case



4.2. Power Planning - Core Power Ring

✓ 4.2. Core Power Ring



— Interleaving and Wire group

	Without Wire group	With Wire group
Without Interleaving	Without wire groups	With wire groups
Interleaving	Without wire groups	With wire groups

4.2. Power Planning - Core Power Ring

import
floorplan
powerplan
placement
CTS
routing

Design Edit Synthesis Partition Floorplan **Power** Place Clock Route Timing SI Verify Tools

1

Power Planning

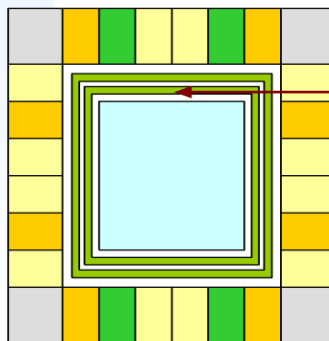
Add Rings...

- In **Basic** tab

Field	Fill In
Top and Bottom Layer	Metal 3
Left and Right Layer	Metal 4
Width	All 9 (or the demanded value)
	Click Update
Offset	◆ Center in channel

2

b



Create power ring

2

Add Rings

Basic Advanced Via Generation

Net(s): **GND VDD**

Ring Type

- Core ring(s) contouring:
 - ◆ Around core boundary
 - ◇ Along I/O boundary
- ☐ Exclude selected objects
- Block ring(s) around
 - ◆ Each block
 - ◆ Each reef
 - ◆ Selected power domain/fences/reefs
 - ◆ Each selected block and/or group of core rows
 - ◆ Clusters of selected blocks and/or groups of core rows
 - ☐ With shared ring edges
- User defined coordinates: MouseClick
- Core ring
 - ◆ Core ring
 - ◇ Block ring

Ring Configuration

	Top:	Bottom:	Left:	Right:
Layer:	met5 H	met5 H	met4 V	met4 V
Width:	9	9	9	9
Spacing:	0.28	0.28	0.28	0.28
Offset:	◆ Center in channel ◇ Specify 0.66 0.66 0.66 0.66			

Option Set

☐ Use option set: Update Basic

OK Variables Apply Defaults Cancel Help

2

Fill in GND VDD

2

c

Update



4.2. Power Planning

- Core Power Ring

— In **Advanced** tab

- You can create one more set of power ring

Field	Fill In
Wire Group	◆ Use wire group ◆ Use wire group
Number of bits	The demanded value

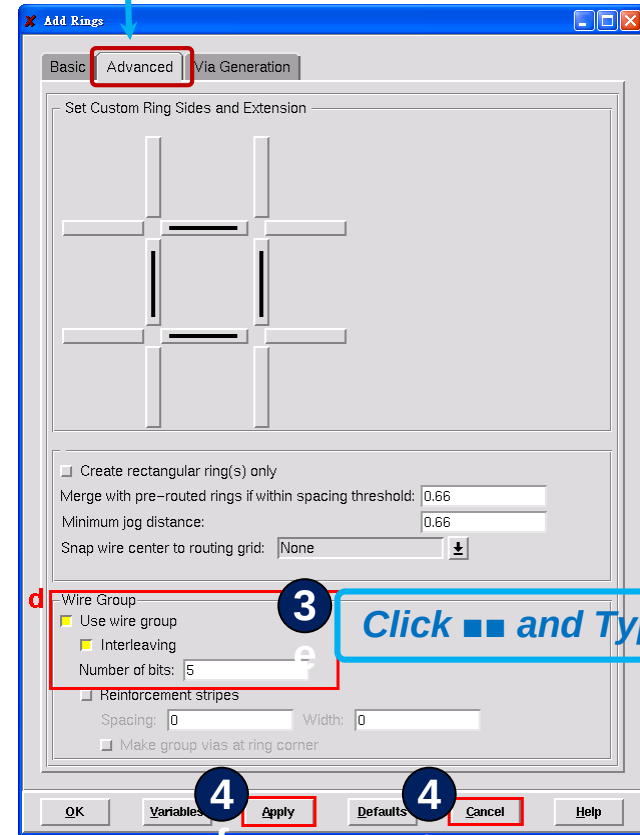
Note:

you can choose the option “Interleaving” to observe the difference in power ring created

- Click [Apply](#)

— Tips

- Click [Apply](#) then you can undo this step
- Click [OK](#) then undo is not allowed



Click [Interleaving checkbox] and Type in num.

4.3. Power Planning - Core Power Pin

import

floorplan

powerplan

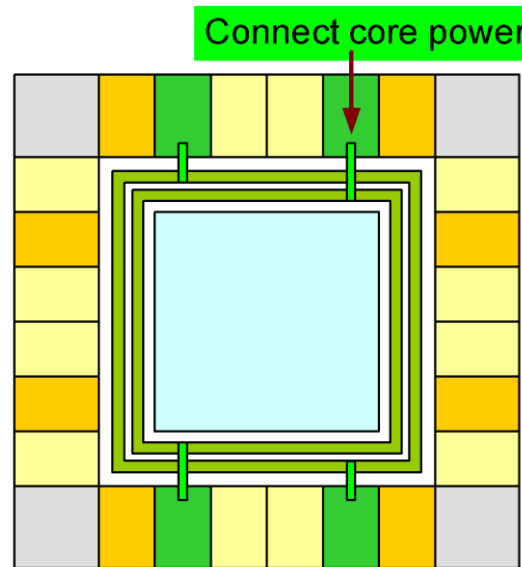
placement

CTS

routing

✓ 4.3. Core Power Pin

- Connect from core power pads to power rings



- Note:
If global nets aren't correctly connected, core power pins will have trouble in generating.

4.3. Power Planning

- Core Power Pin

import
floorplan
powerplan
placement
CTS
routing

Design Edit Synthesis Partition Floorplan Power Place Clock **Route** Timing SI Verify Tools

1

Special Route ...

✓ Connect core power pin

- Complete the form and click [Apply](#)

Field	Fill In
Block pins	◇ (off)
Pad pins	◆ (on)
Pad rings	◇ (off)
Standard cell pins	◇ (off)
Stripes (unconnected)	◇ (off)

2

Choose M2 for Bottom layer (Suggestion)

Note:

Make sure that core power pads are connected to the power ring.
Adjust the top and bottom layer if needed.

2

Basic Advanced Via Generation

Net(s): GND YDD

Route

☐ Block pins ☐ Pad pins ☐ Pad rings ☐ Standard cell pins ☐ Strip es (unconnected)

Level shifter pins

Net(s):

Stripe Layer: M6 Width: Pitch:

Routing Control

Layer Change Control

Top layer: M6 Bottom layer: M1

☐ Straight connections and allow jogging ☐ Straight connections only ☐ Same layer routing only

☐ Prefer straight with layer change ☐ DR C clean

☐ Prefer different layer jog ☐ Allow layer change

☐ Prefer same layer jog

Area Draw

X1: Y1:

X2: Y2:

☐ Connect to target inside the area only

☐ Delete existing routes

☐ Generate progress messages 0

☐ Extra config file:

OK Apply Defaults Cancel Help

2



4. Power Planning – Stripes

import

floorplan

powerplan

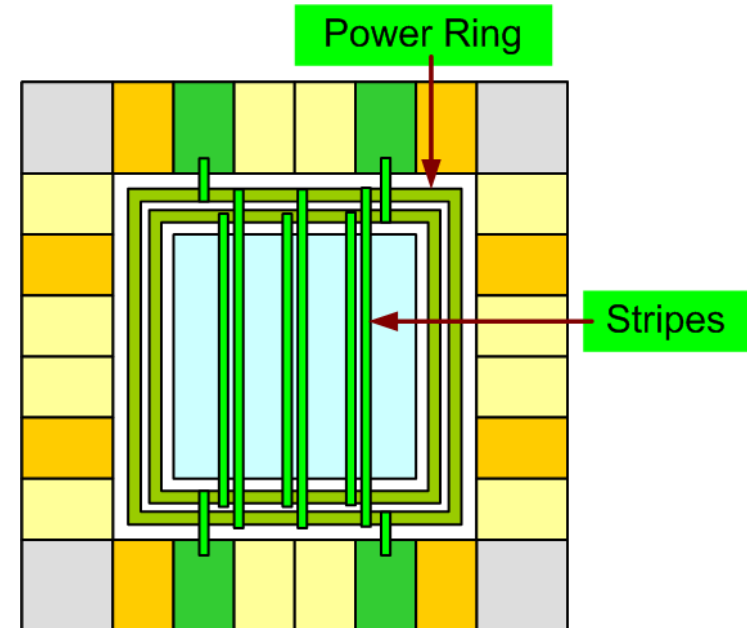
placement

CTS

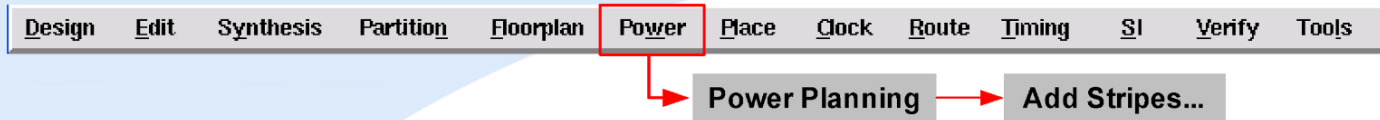
routing

✓ Calculate stripe set

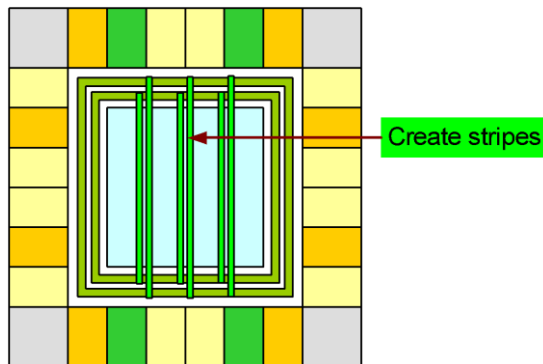
- Experience
 - Add one stripe set per 100 μm
- Example
 - Core width = height = 1600
 - Stripe set added = 15



4.4. Power Planning - Stripes



- Complete the form in Basic tab and click [Apply](#)



Note: after click point

Start (X): click the location of first stripe in design display area

Stop (X): click the location of last stripe in design display area

Note: Specify Start (Y) and Stop (Y) for horizontal stripes

Add Stripes

Basic Advanced Via Generation

Set Configuration

Net(s): GND VDD

Layer: me12

Direction: Vertical Horizontal

Width: 9

Spacing: 0.28

Update

Set Pattern

Set-to-set distance: 100

Number of sets: 3

Bumps: Over Between

Over PPG pins Pin layer: Top pin layer Max pin width: 0

Master name: Selected blocks All blocks

Stripe Boundary

Core ring

Pad ring Inner Outer

Design boundary Create pins

Each selected block/domain/fence

All domains

Specify area

First / Last Stripe

Start from: left right

Relative from core or selected area

Absolute locations

Start (X): 336.131 Point... Stop (X): 896.666 Point...

Option Set

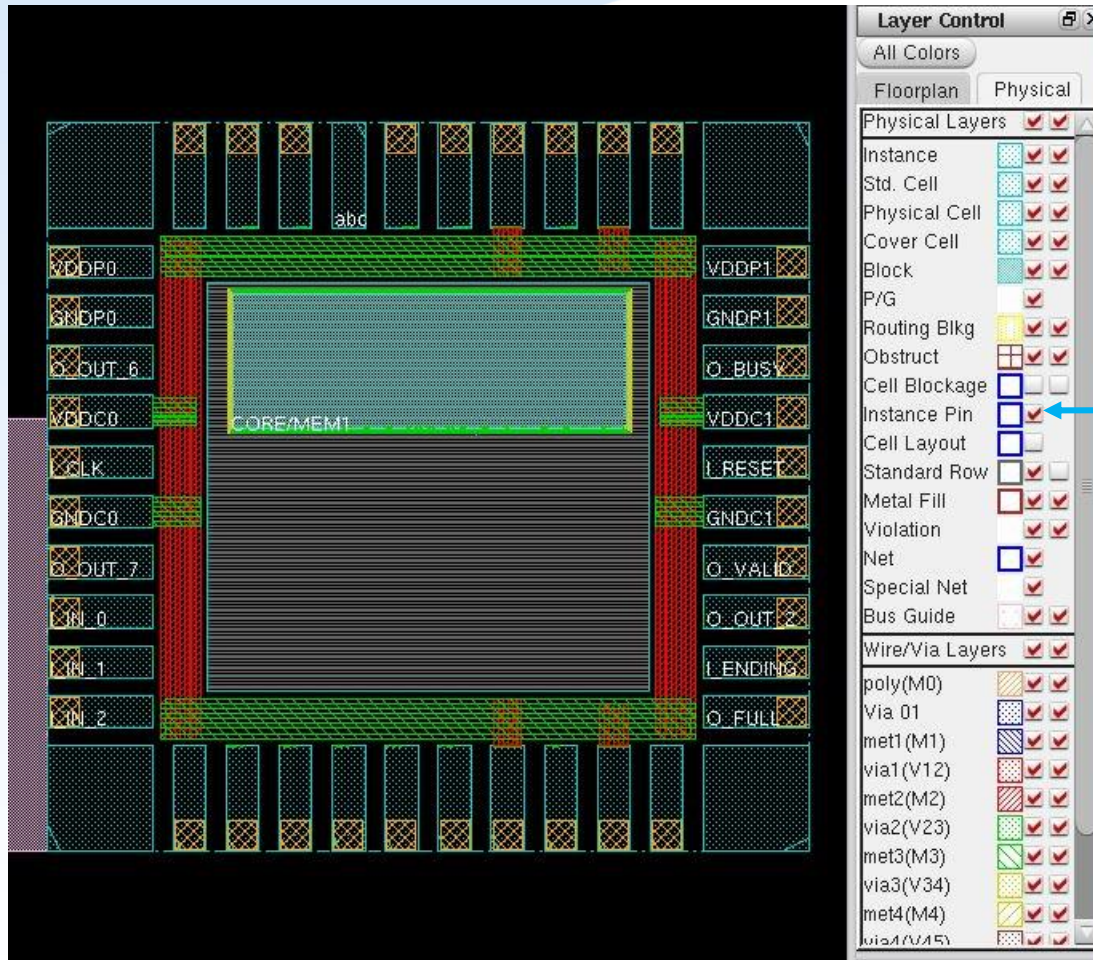
Use option set: Update Basic

OK Variables **Apply** Defaults Cancel Help

4. Power Planning

– Block Ring for Hard Macro

import
floorplan
powerplan
placement
CTS
routing



*To visualize the
block ring of
SRAM*

4. Power Planning

– Block Ring for Hard Macro

import

floorplan

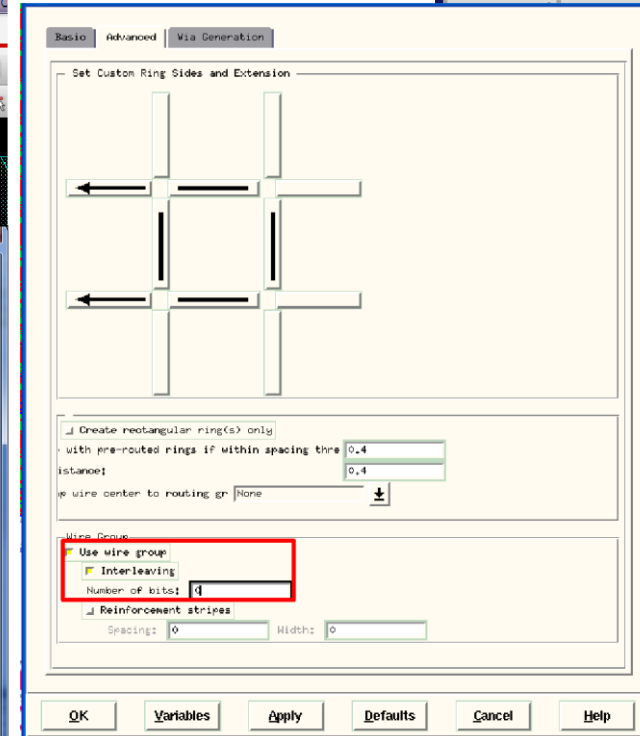
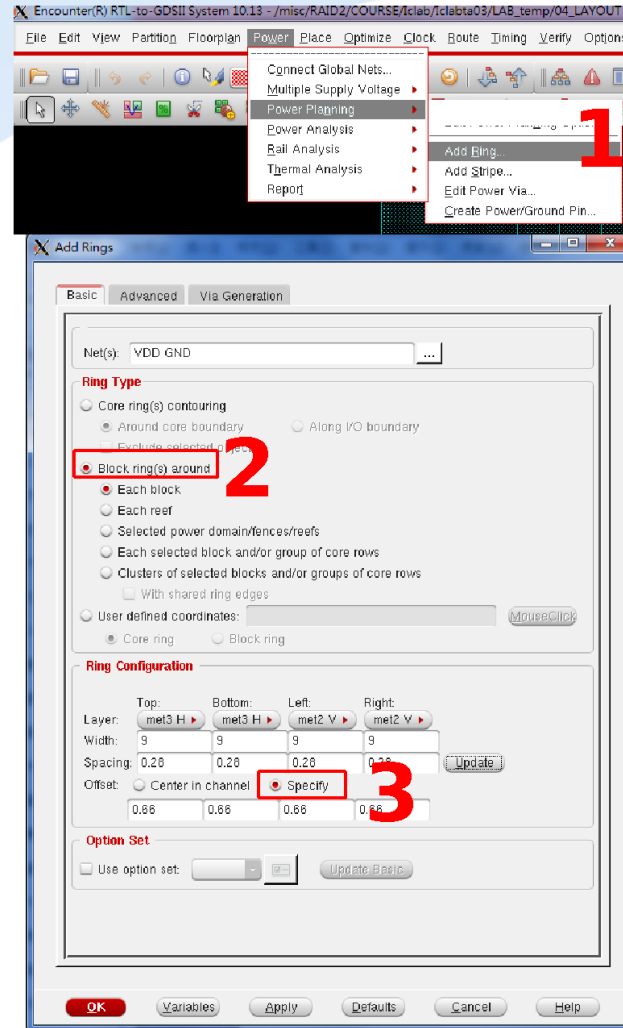
powerplan

placement

CTS

✓ Add block ring

- Specify nets
- Block ring around
- Specify metals
- Width, offset
- Extension
- Wiregroups



4.5. Standard Cell Power Connection

import

floorplan

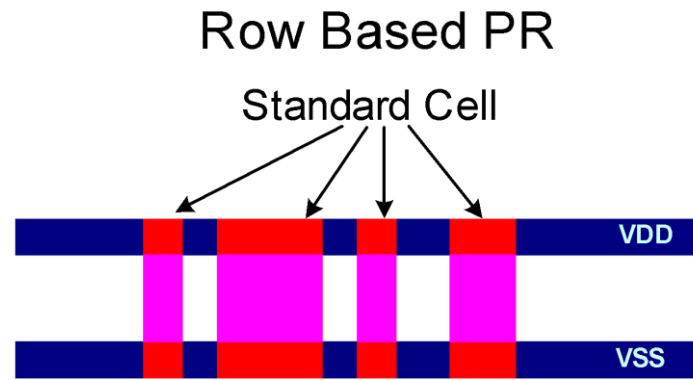
powerplan

placement

CTS

routing

✓ Connect standard cell power line



– Text Command

```
innovus #> sroute -noBlockPins -noPadRings -noPadPins -noStripes -jogControl  
{ preferWithChanges differentLayer }
```



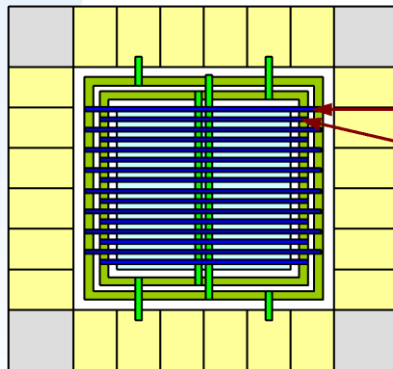
4.5. Standard Cell Power Connection

✓ Connect standard cell power line

Design Edit Synthesis Partition Floorplan Power Place Clock **Route** Timing SI Verify Tools

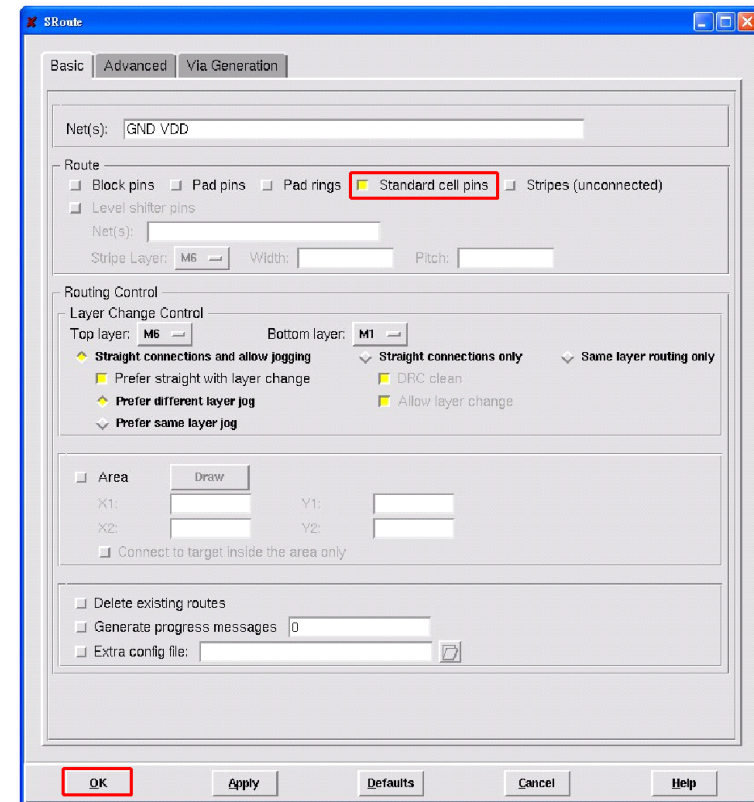
Special Route ...

Field	Fill In
Block pins	◇ (off)
Pad pins	◇ (off)
Pad rings	◇ (off)
Standard cell pins	◆ (on)
Stripes (unconnected)	◇ (off)



Std. cell power line

Std. cell ground line



– Text Command

innovus #> **sroute -noBlockPins -noPadRings -noPadPins -noStripes -jogControl
{ preferWithChanges differentLayer }**

4.6 Add PAD Filler

✓ Adding pad filler

- There should be no spacing between pads.
Therefore adding pad filler is necessary for core limited design
- PAD filler must be added before detail route, or there may have some DRC/LVS violations after PAD filler insertion



– Text Command

From wider fillers to narrower ones

- innovus #> addIoFiller -cell PFILL -prefix IOFILLER
- innovus #> addIoFiller -cell PFILL 9 -prefix IOFILLER
- innovus #> addIoFiller -cell PFILL 1 -prefix IOFILLER
- innovus #> addIoFiller -cell PFILL 01 -prefix IOFILLER -fillAnyGap

Only the smallest filler can use -fillAnyGap option



5. Standard Cell Placement

import

floorplan

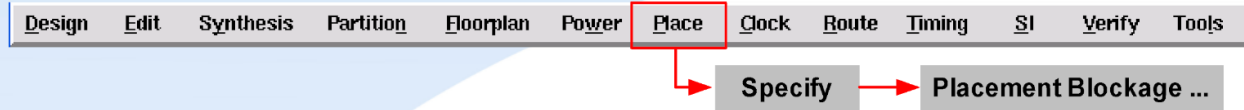
powerplan

placement

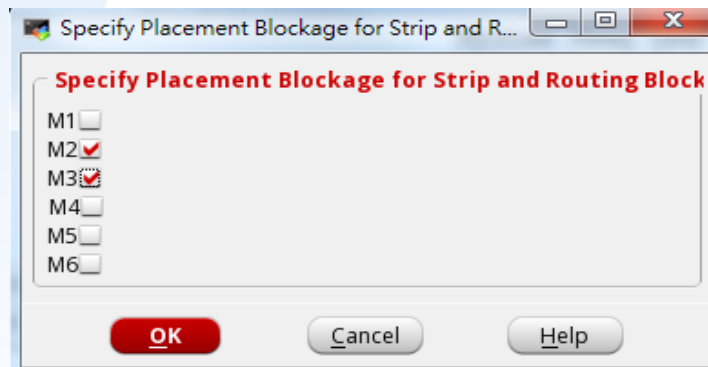
CTS

routing

✓ Specify placement blockage



- Choose the metal layers of stripe.
Then there will be no cell placed under stripe
- Text Command
innovus #> setPrerouteAsObs {2 3}



- Add placement blockage for macros
- Text Command
innovus #> createObstruct llx lly urx ury



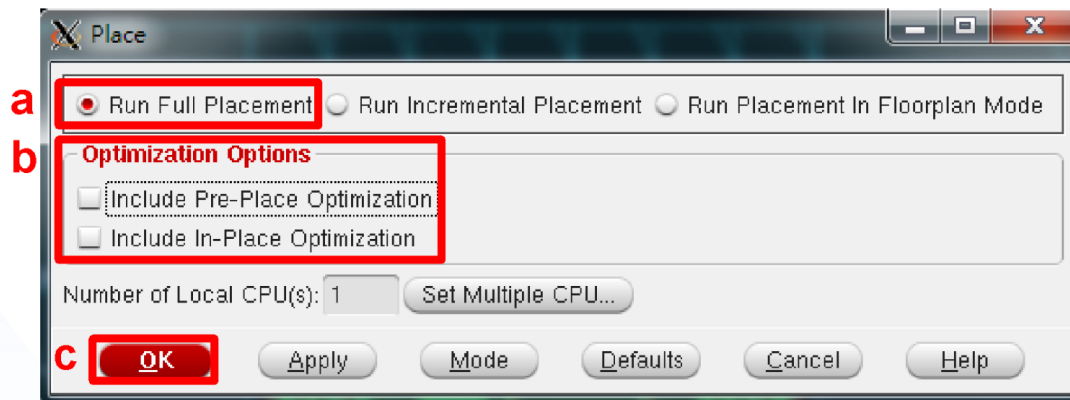
5. Standard Cell Placement

✓ Place standard cells



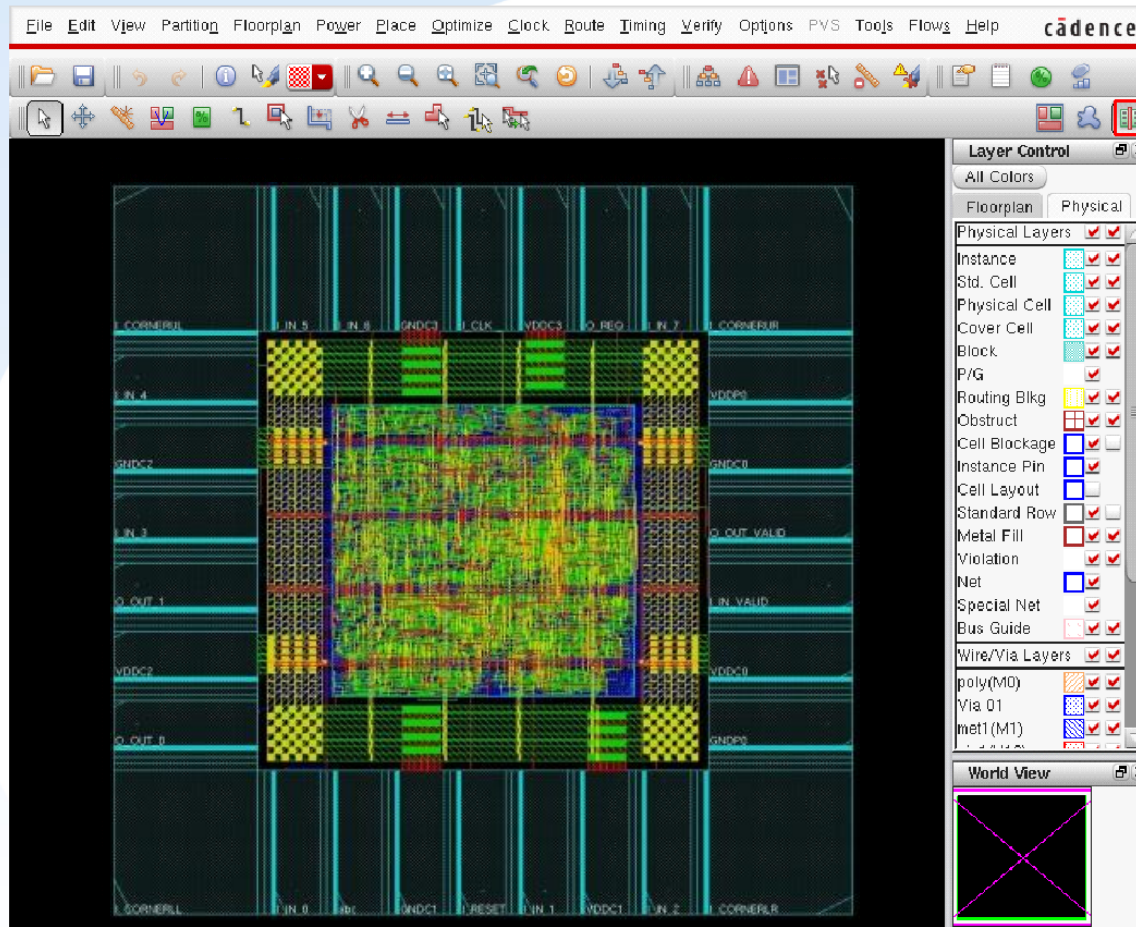
Standard Cells And Blocks...

- Run Full Placement
- Optimization Options
 - ☐ Include Pre-Place Optimization
 - ☐ Include In-Place Optimization
- Click OK button



5. Standard Cell Placement – Result

import
floorplan
powerplan
placement
CTS
routing



6. Pre-CTS Timing Analysis

✓ To check timing

- innovus #> timeDesign -preCTS

✓ Operations

- Trial route
- RC extraction
- Timing analysis
- Generates reports are saved in ./timingReports/ , including
 - _preCTS.cap
 - _preCTS.fanout
 - _preCTS.tran
 - _preCTS_all.tarpt

✓ Reports

- Congestion distribution
- Timing Summary
- If the timing is met, pre-CTS optimization can be skipped

Congestion distribution:

Remain	cntH		cntV	
-2:	5	0.02%	0	0.00%
-1:	10	0.03%	0	0.00%
1:	10	0.03%	21	0.06%
2:	10	0.03%	58	0.17%
3:	0	0.00%	88	0.25%
4:	33	0.10%	379	1.09%
5:	0	0.00%	4227	12.18%

Be careful of negative congestion

Setup mode	all	reg2reg	in2reg	reg2out	in2out	clkgate
WNS (ns):	-0.744	N/A	-0.744	N/A	N/A	N/A
TNS (ns):	-7.291	N/A	-7.291	N/A	N/A	N/A
Violating Paths:	16	N/A	16	N/A	N/A	N/A
All Paths:	40	N/A	40	N/A	N/A	N/A

Negative slack should be fixed

Density: 19.855%

Routing Overflow: 0.00% H and 0.00% V

Real DRV (fanout, cap, tran): (0, 1, 0)

Total DRV (fanout, cap, tran): (0, 1, 0)

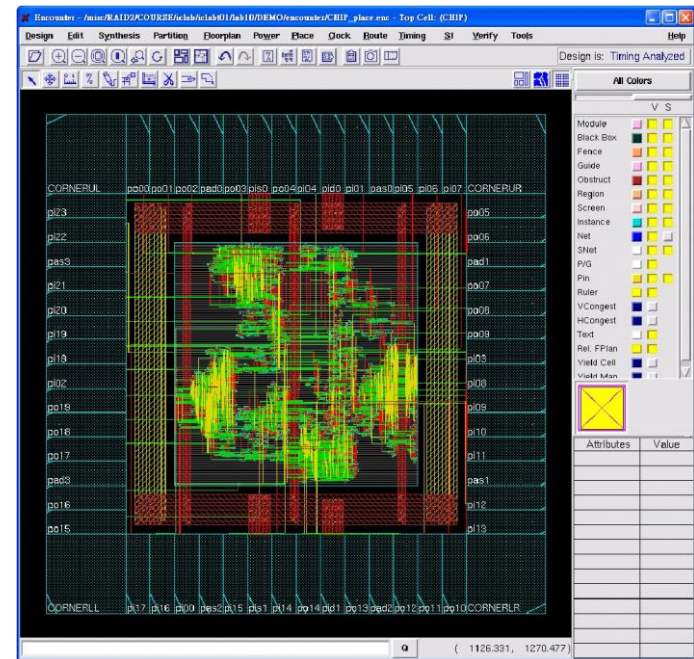
Total DRV = real DRV + clock network DRV

6. Pre-CTS Optimization

- ✓ To optimize timing placed design for the first time with ideal clocks
 - innovus #> optDesign -preCTS
- ✓ To further optimize a design after above command execution
 - innovus #> optDesign -preCTS -incr
- ✓ Operation
 - Repair Setup slack, Setup times
 - Repair Design rule violations (DRVs) and remaining DRVs

Setup mode	all	reg2reg	in2reg	reg2out	in2out	clkgate
WNS (ns):	0.102	N/A	0.102	N/A	N/A	N/A
TNS (ns):	0.000	N/A	0.000	N/A	N/A	N/A
Violating Paths:	0	N/A	0	N/A	N/A	N/A
All Paths:	40	N/A	40	N/A	N/A	N/A

Density: 20.928%
Routing Overflow: 0.00% H and 0.00% V
Real DRV (fanout, cap, tran): (0, 0, 0)
Total DRV (fanout, cap, tran): (0, 0, 0)



7. Clock Tree Synthesis - Clock Problem

import

floorplan

powerplan

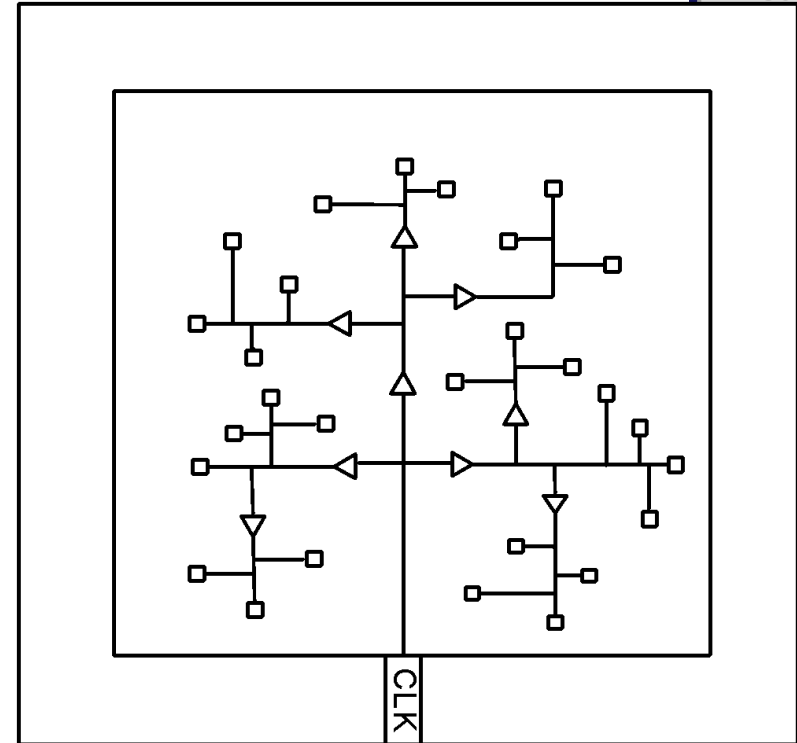
placement

CTS

routing

✓ Clock problem

- Heavy clock net loading
- Long clock insertion delay
- Clock skew
- Skew across clocks
- Clock to signal coupling effect
- Clock is power hungry
- Electromigration on clock net



✓ Clock is one of the most important treasure in a chip, do not take it as other use.

7. Clock Tree Synthesis

import

floorplan

powerplan

placement

CTS

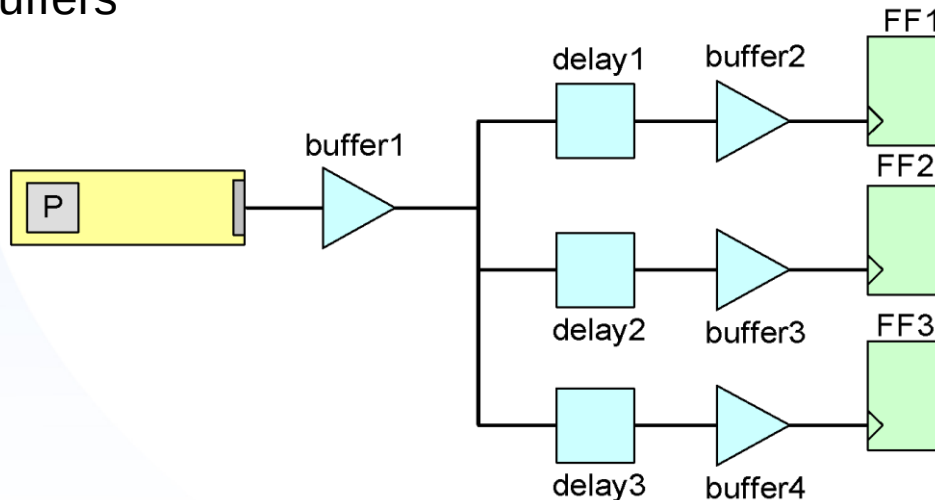
routing

✓ The goal of clock tree synthesis includes

- Creating clock tree spec file
- Building a buffer distribution network
- Routing clock nets using CTS-NanoRoute

✓ In automatic CTS mode, Innovus will do the following things

- Build the clock buffer tree according to the clock tree specification file
- Balance the clock phase delay with appropriately sized, inserted clock buffers



8. Post-CTS Timing Analysis - Text Command

✓ To check setup time

- innovus #> **timeDesign** –postCTS

✓ To check hold time

- innovus #> **timeDesign** –postCTS –hold

✓ Reports

- Generated timing reports are saved in ./timingReports/, including _postCTS.cap , _postCTS.fanout , _postCTS.tran , _postCTS_all.tarpt”

✓ To correct setup violations and design rule violation

- innovus #> **optDesign** –postCTS

✓ To correct hold violations

- innovus #> **optDesign** –postCTS –hold

✓ Operation

- Repair Setup slack, Setup times
- Repair Design rule violations (DRVs)



10. NanoRoute – Intro.

import

floorplan

powerplan

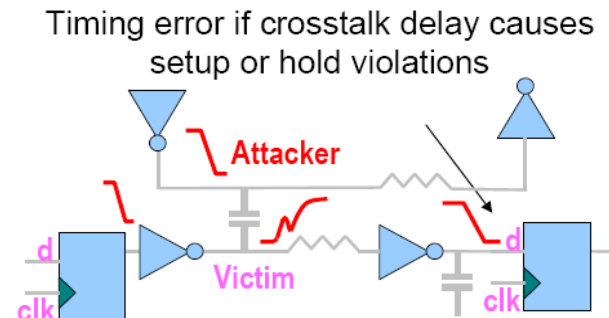
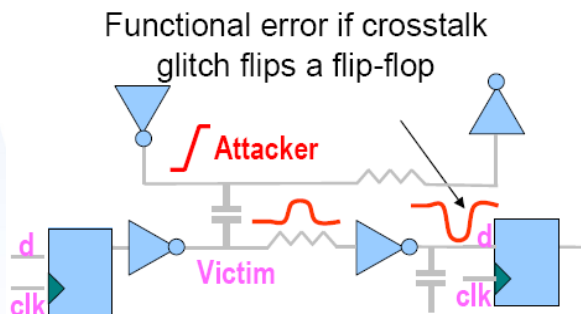
placement

CTS

routing

✓ Signal Integrity (SI) Issue

- Crosstalk
- Charge sharing
- Supply noise
- Leakage
- Propagated noise
- Overshoot
- Under shoot



✓ SI closure

- Occur when a design is free from SI induced functional glitch and timing failure



Appendix: Crosstalk

import

floorplan

powerplan

placement

CTS

routing

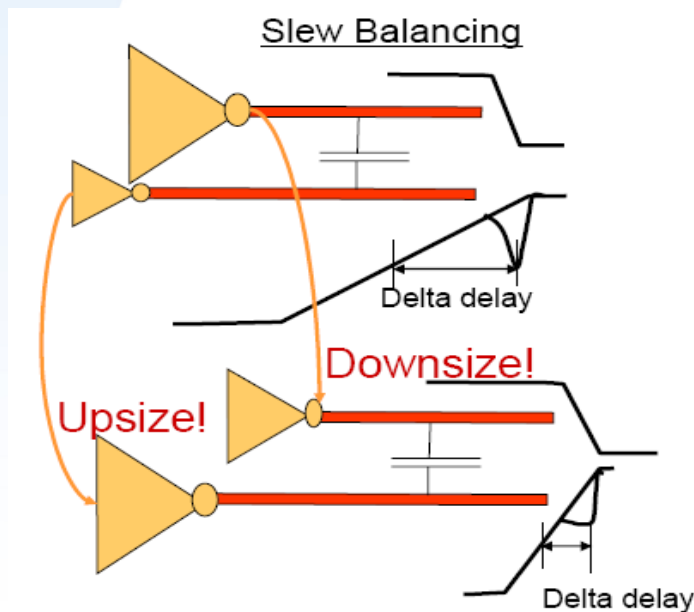


✓ Crosstalk prevention – Placement solution

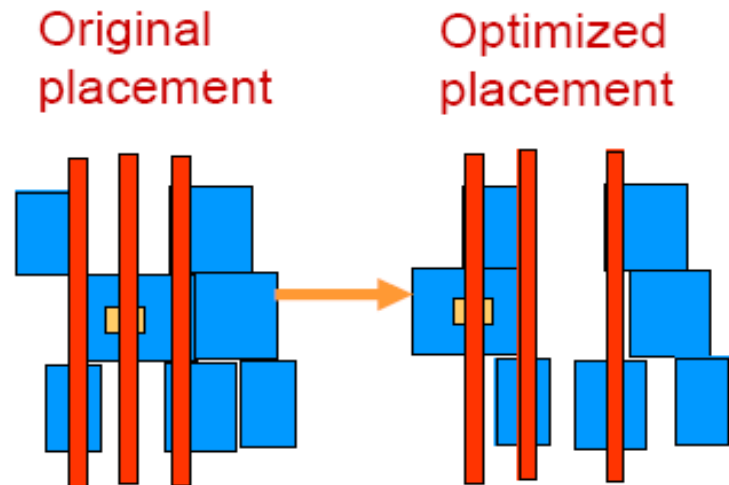
- Insert buffer in lines
- Upsize driver
- Congestion optimization

✓ Crosstalk prevention – Routing solution

- Limit length of parallel nets
- Wider routing grid
- Shield special nets



- Reduce coupling capacitance



10. NanoRoute – Intro.

import

floorplan

powerplan

placement

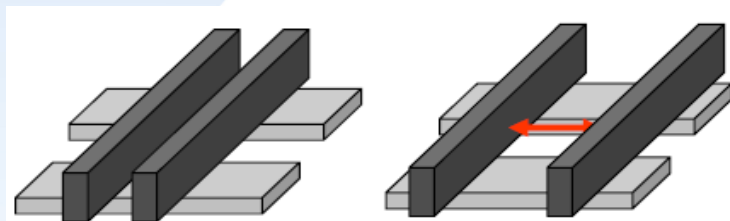
CTS

routing

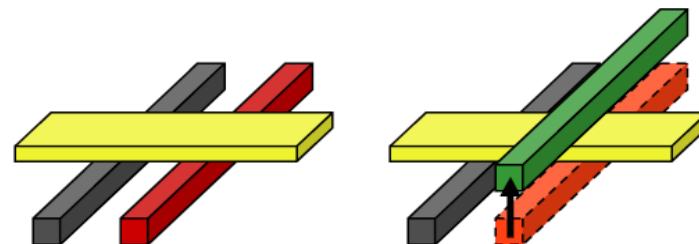
✓ SI prevention (cont.)

– Routing-based SI prevention

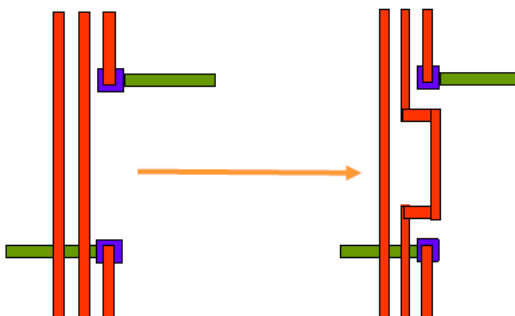
• Wiring Spacing



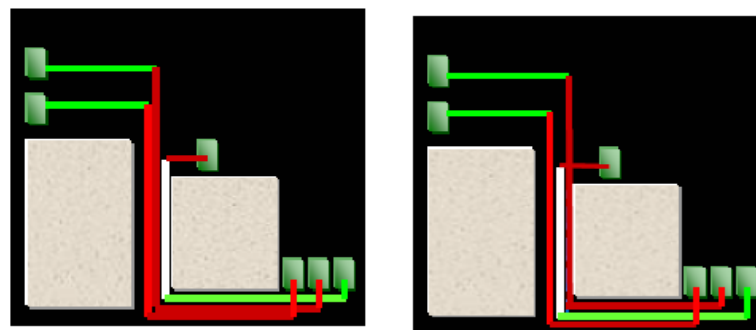
• Layer Switching



• Parallel Wires Reducing



• Net Re-ordering



10. NanoRoute – Process Antenna Effect

import

floorplan

powerplan

placement

CTS

routing

✓ Antenna Effect

- In a chip manufacturing process, metal is initially deposited so it covers the entire chip
- Then, the unneeded portions of the metal are removed by etching, typically in plasma (charged particles).
- The exposed metal collect charge from plasma and form voltage potential.
- If the voltage potential across the gate oxide becomes large enough, the current can damage the gate oxide.

✓ **Antenna Ratio = $\frac{\text{Area of process antennas on a node}}{\text{Area of gates to the node}}$**

✓ Problem Repair

- Add jumper (changer metal layer)
- Add antenna cell (diode)
- Add dummy transistor



10. NanoRoute – Process Antenna Effect

import

floorplan

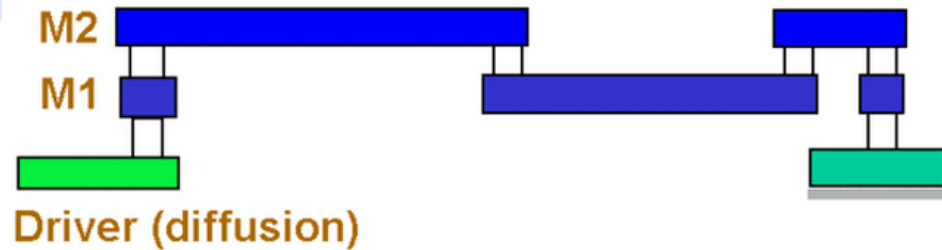
powerplan

placement

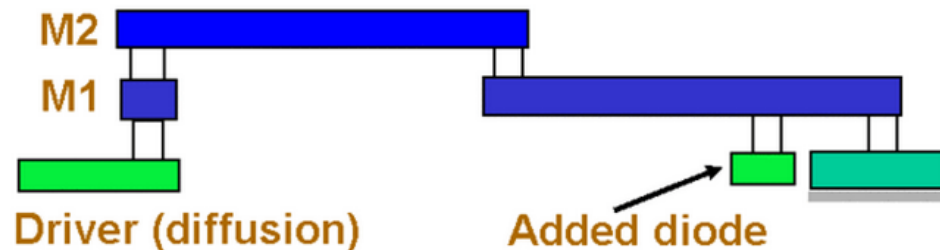
CTS

routing

✓ Add jumper



✓ Add diode



✓ If after fixing antenna violation, there still exist violations

- Larger chip area
- Place module precisely



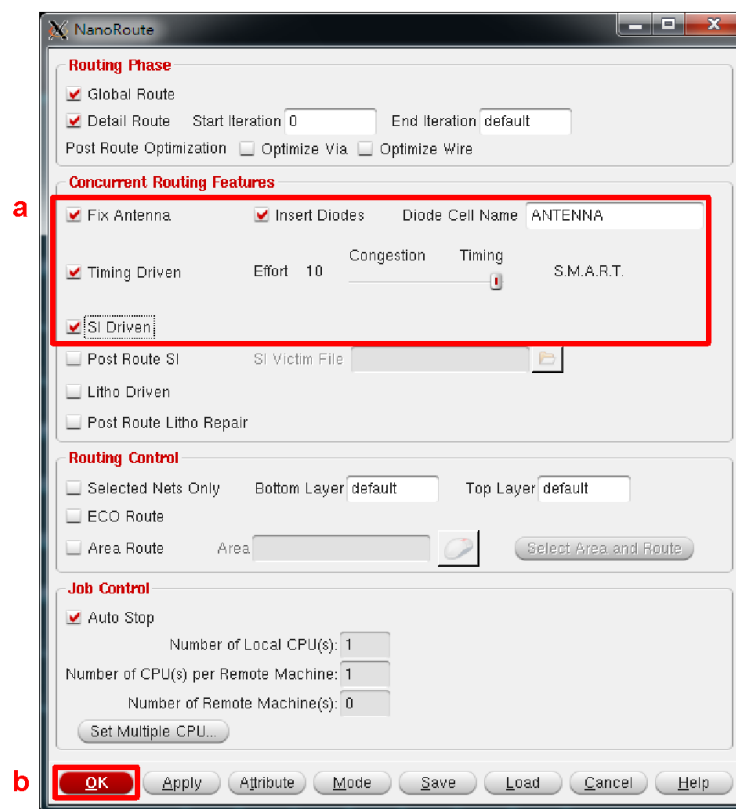
10. NanoRoute – GUI

✓ NanoRoute

- Routing the design without creating DRC or LVS violations
- Routing the design without degrading timing or creating signal integrity violations
- Complete the form and click OK

Field	Fill In
Concurrent Routing Features	◆ Fix Antenna
	◆ Insert Diodes
	Diode Cell Name ANTENNA
	◆ Timing Driven
	◆ SI Driven

Add I/O Pad Filler before Routing!!



11. Post-Route Timing Analysis

import

floorplan

powerplan

placement

CTS

routing

✓ To check setup time

- innovus #> timeDesign –postRoute

✓ To check hold time

- innovus #> timeDesign –postRoute –hold

✓ Operation

- Repair Setup slack, Setup times
- Repair Design rule violations (DRVs) and remaining DRVs

✓ Reports

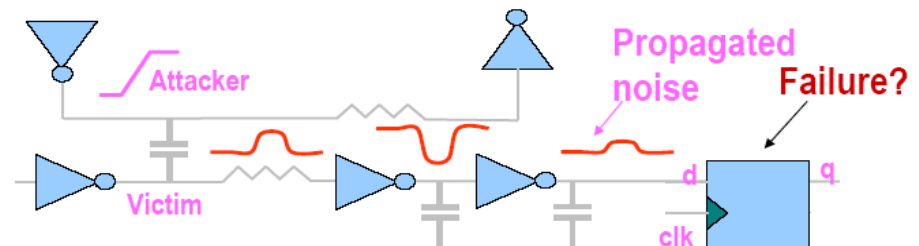
- Generated timing reports are saved in ./timingReports/ , including
 - postRoute.cap
 - _postRoute.fanout
 - _postRoute.tran
 - _postRoute_all.tarpt



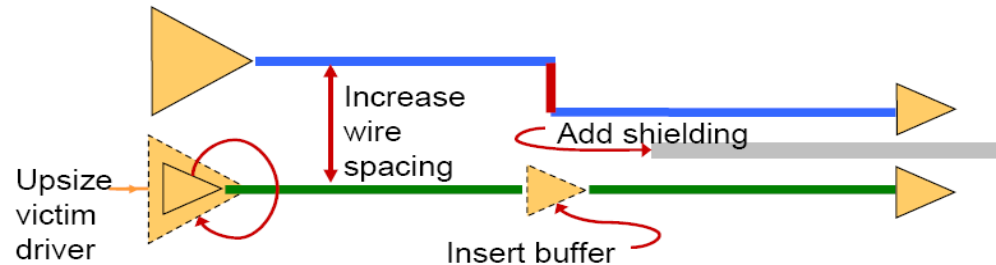
12. Post-Route Timing Analysis (include SI)

✓ Text Command

- To do glitch noise analysis
 - innovus #> **timeDesign –postRoute –si**



✓ SI repair techniques



✓ Text command

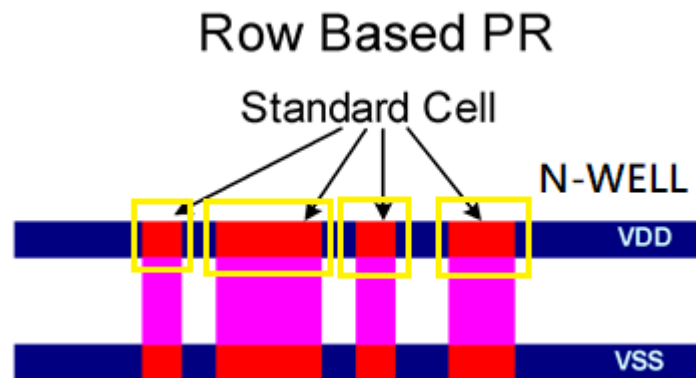
- To correct setup violations and design rule violations
 - innovus #> **optDesign –postRoute –si**
- To correct hold violations
 - innovus #> **optDesign –postRoute –si –hold**

13. Add Filler Cells

✓ Purpose of adding filler cells

- Fill all the gaps between standard cell instances
- Provide decoupling capacitances to complete connections in the standard cell rows

✓ Metal filler inserted after routing, but before parasitic extraction and GDSII out



13. Add Filler Cells

✓ Add filler cell into the free space of core design

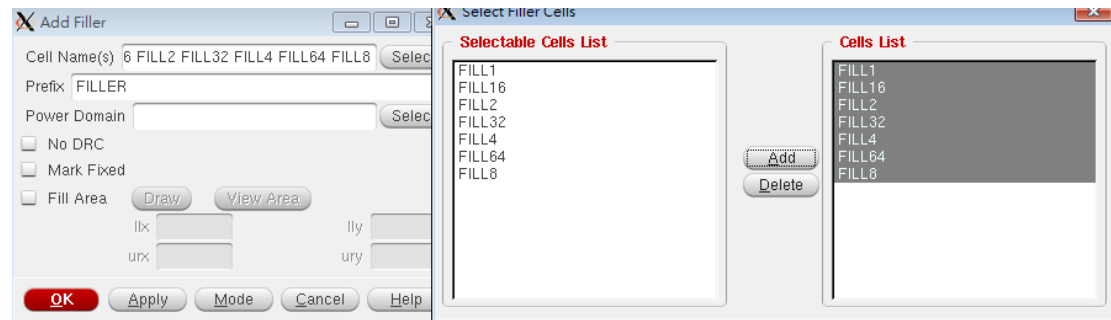
Design Edit Synthesis Partition Floorplan Power **Place** Clock Route Timing SI Verify Tools

Physical Cell

Add Filler

Why add dummy

- *meet minimize metal density rule*
- *prevent over etching*
- *prevent sagging in local area*
- *improve yield*
- *reduce on chip variation*



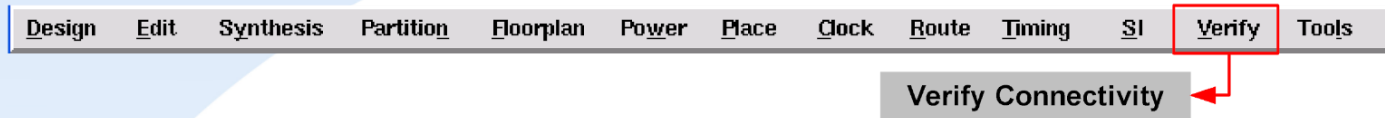
✓ Text command

- innovus #> **addFiller -cell FILL64 -prefix filler64**
- innovus #> **addFiller -cell FILL32 -prefix filler32**
- innovus #> **addFiller -cell FILL16 -prefix filler16**
- innovus #> **addFiller -cell FILL8 -prefix filler8**
- innovus #> **addFiller -cell FILL4 -prefix filler4**
- innovus #> **addFiller -cell FILL2 -prefix filler2**
- innovus #> **addFiller -cell FILL1 -prefix filler1**



14. Before Output: LVS & DRC check

✓ LVS(layout verse schemectic)



✓ DRC(design rule check)



✓ Reminder

- Recommended to perform the two checks after power planning
- Check your design before streaming out

14. LVS v.s. DRC v.s. DRV

- ✓ **LVS: layout verse schematic**
 - The signals are connect correctly
- ✓ **DRC: design rule check**
 - Metal density
 - Size of poly, metals, vias
- ✓ **DRV: design rule violation**
 - Fan-out
 - Load
 - Driving capability
 - Wire length



14. Output Files

✓ The following files will be generated after APR design flow

- **CHIP.v**
 - Netlist file for post-layout gate-level simulation
- **CHIP.sdf**
 - Design timing file for post-layout gate-level simulation
- **CHIP.def**
 - Design exchange format relevant to physical layout
- **CHIP.gds**
 - GDSII stream file for Calibre-DRC (Design Rule Check)



14. Output Data

✓ Export Netlist for LVS and simulation

- File -> Save -> Netlist ...
- innovus #> `saveNetlist CHIP_LAYOUT.v`
- innovus #> `saveNetlist -includePhysicalInst -excludeLeafCell
CHIP_LVS.v`

✓ Save sdf for post layout simulation

- Innovus #> `setAnalysisMode -checkType setup -analysisType bcwc -
cpr none -clockGatingCheck 1 -timeBorrowing 1 -domain clock -
useOutputPinCap 1`
- innovus #> `write_sdf -edges check_edge CHIP.sdf`



14. Output Data

- ✓ Export GDSII stream file for DRC, LVS, LPE (Layout Parameter Extration), and tape out
 - CHIP.gds
- ✓ Merging GDSII files (UMC 0.18)
 - Merge the GDSII files of standard cells (and macros if any) into complete layout and output to a single GDSII file for hierarchical designs

Design Edit Synthesis Partition Floorplan Power Place Clock Route Timing SI Verify Tools

Save → GDS...

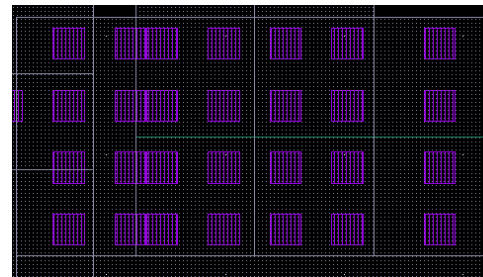
Field	Fill In
Output Stream File	CHIP.gds
Map File	streamOut.map
Library Name	CHIP
◆ GDS Structure Name	CHIP
◆ Merge Stream Files	umc18.gds2 umc18io3v5v_6lm.gds2

Appendix: More on APR

✓ DRC & LVS in Virtuoso with streamout GDS

✓ DRC Result

- PO.R.3
- M2.R.1
- VIA3.S.1



OVERALL COMPARISON RESULTS

```
#
#
# #
# #
#
#####
#
# CORRECT
#
#####
```

```
* *
* I *
* _ *
```

Warning: Ambiguity points were found and resolved arbitrarily.

```
*****
CELL SUMMARY
*****
```

Result	Layout	Source
-----	-----	-----
CORRECT	CHIP	CHIP

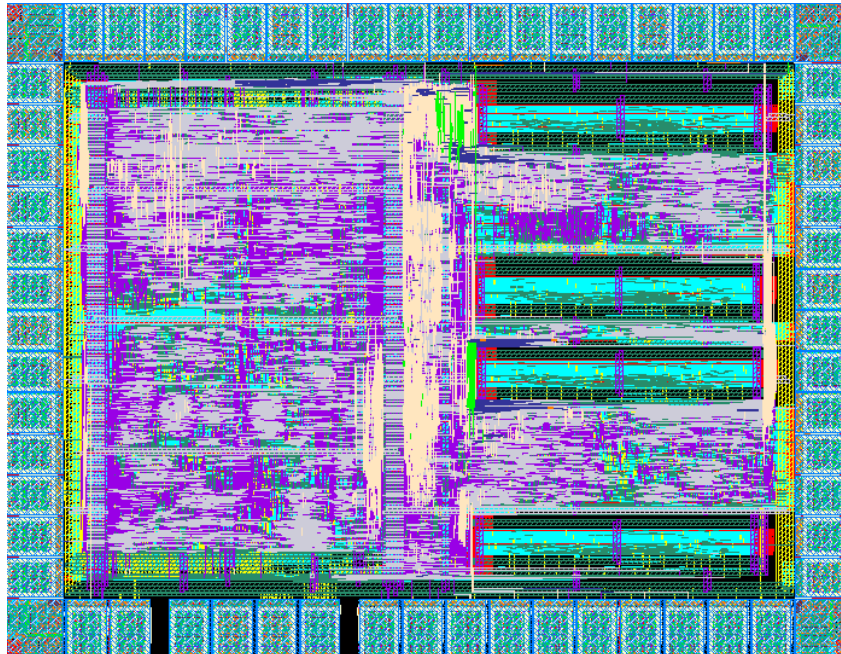


Appendix: More on APR

✓ Macro Layout

- Pad -> Pin
- No Shell

✓ Different Power Domain



Appendix

✓ Modify the Memory.lef file

- ❖ Change “core” to “core_5040”
- ❖ Change “ME4” to “metal4” and so on
- ❖ Change “VI1” to “via”
- ❖ Change “VI2” to “via2” and so on

```
RECT 0.000 0.140 213.900 294.000 ;  
LAYER VI1 ;  
RECT 0.000 0.140 213.900 294.000 ;  
LAYER VI2 ;  
RECT 0.000 0.140 213.900 294.000 ;  
LAYER VI3 ;  
RECT 0.000 0.140 213.900 294.000 ;
```

```
NAMESCASESENSITIVE ON ;  
MACRO TEST  
CLASS BLOCK ;  
FOREIGN TEST 0.000 0.000 ;  
ORIGIN 0.000 0.000 ;  
SIZE 213.900 BY 294.000 ;  
SYMMETRY x y r90 ;  
SITE core ;  
PIN VCC  
DIRECTION INOUT ;  
USE POWER ;  
SHAPE ABUTMENT ;  
PORT  
LAYER ME4 ;  
RECT 212.780 282.580 213.900 285.820 ;  
LAYER ME3 ;  
RECT 212.780 282.580 213.900 285.820 ;  
LAYER ME2 ;  
RECT 212.780 282.580 213.900 285.820 ;  
LAYER ME1 ;  
RECT 212.780 282.580 213.900 285.820 ;  
END
```

